

RIGID ANALYTIC EISENSTEIN COCYCLES FOR $\mathrm{SL}_n(\mathbb{Z}[1/p])$

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ABSTRACT. We define *rigid analytic Eisenstein cocycles* for (congruence subgroups of) the p -arithmetic groups $\mathrm{SL}_n(\mathbb{Z}[1/p])$, generalizing the Dedekind–Rademacher construction of Darmon–Pozzi–Vonk when $n = 2$ to arbitrary dimension and arbitrary conductor. We show that evaluations of these cocycles at special moduli yield Gross–Stark units in abelian extensions of totally real fields in which p is inert, using the Gross tower of fields conjecture proven by Dasgupta–Kakde.

Our construction proceeds through cocycles valued in integral adelic distributions, obtained from an explicit differential form parametrization of toric polylogarithms via the geometry of S^{n-1} . We identify specializations of these cocycles with L -values of abelian extensions of totally real fields and use the proof of Gross’s tower of fields conjecture to deduce the unit formula. To establish the comparison with L -values, we construct a Hodge refinement of the regularized Eisenstein kernel of Bergeron–Charollois–Garcia in the toric setting. This refinement represents a lift of their polylogarithm class to coherent cohomology, following ideas of Kings–Sprang.

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1. INTRODUCTION

If F is an imaginary quadratic field, then the theory of modular functions for subgroups of $\mathrm{SL}_2(\mathbb{Z})$ acting on the complex upper-half plane $\mathcal{H}$ offers an explicit analytic approach to understanding abelian extensions of F , or in other words a solution to Hilbert’s twelfth problem for imaginary quadratic fields. Namely, there exist certain invertible holomorphic functions

$${}_c g_{\alpha,\beta} \in \mathcal{O}_{\mathcal{H}}^\times,$$

called modular (or Siegel) units, invariant under arithmetic subgroups of $\mathrm{SL}_2(\mathbb{Z})$, such that the values ${}_c g_{\alpha,\beta}(\tau)$ at imaginary quadratic points $\tau \in \mathcal{H}$ lying in F generate such extensions; these values are called *elliptic units* for the field, and have rich internal arithmetic structure.

There is no naive analogue of this theory for *real* quadratic fields F since $\mathcal{H}$ does not contain real quadratic points. However, Darmon and Dasgupta [DD] proposed a conjectural construction of elliptic units for real quadratic fields using a p -adic limiting process involving periods of logarithmic derivatives of modular units along real quadratic geodesics, and conjectured that they enjoyed analogous properties as classical elliptic units (above all, being algebraic and generating abelian extensions). Dasgupta later suggested an alternate approach to constructing conjectural elements in these extensions, using p -adic integrals against distributions constructed from a refinement of Shintani’s method for calculating L -values of totally real fields. Using this, he gave conjectural p -adic analytic formulas for p -units in abelian extensions of general totally real fields F [Das08]. Recently, the remarkable work of Dasgupta and Kakde confirmed this conjecture; more precisely, they proved that their resulting objects are precisely the p -units satisfying the conjectures of Gross-Stark and Brumer-Stark [DK23] [Das+23]. This furnishes a complete p -adic analytic solution to explicit class field theory for totally real fields.

The original formula of Dasgupta mentioned above involved an intricate p -adic multiplicative integral, with the measure constructed out of a choice of well-behaved Shintani cone decomposition for the totally real field in question. In later joint work, this approach was organized into the cohomological language of “Eisenstein” or “Shintani” cocycles paired with the fundamental class of the $(n-1)$ -torus associated to $\mathcal{O}_F^\times$ (e.g. with Spiess in [DS18]); the proof of the equivalence of various such approaches was established in Dasgupta–Honnor–Spiess [DH]. To date, such cohomological formulas, even when they can be written in terms of a cocycle for GL_n or SL_n , involve auxiliary data which must be specified individually for different fields (e.g. an irrational “Colmez perturbation” vector to pin down a Shintani decomposition, or a Q -parameter). One of the aims of our construction in this article is to give a simpler cohomological formalism showing that it is possible to achieve a more uniform description.

1.1. Rigid analytic cocycles and arithmetic. A program towards a modular framework for understanding these constructions, more closely analogous to the classical theory of modular units, appeared in the work of Darmon, Pozzi, and Vonk [DPV2]. They consider analogues of modular functions, which they call *rigid analytic cocycles* (or cohomology classes), which can be evaluated at real quadratic points in the p -adic upper half-plane $\mathcal{H}_p$. In particular, in [DPV2], they produce what they term a *rigid analytic Dedekind–Rademacher cocycle* in

$$H^1(\mathrm{SL}_2(\mathbb{Z}[1/p]), \mathcal{A}^\times/p^\mathbb{Z}),$$

where $\mathcal{A}^\times := \mathcal{O}_{\mathcal{H}_p}^\times$ denotes the invertible rigid analytic functions. The evaluation of this class involves a specialization and cap product in addition to the usual evaluation of functions. They then expressed the original construction of [DD] as the value of this rigid class, and gave a new proof of the conjecture of [DD] in this setting.

More broadly, rigid analytic cocycles for SL_2 were given a structure theory in [DV], and are hoped to fit into a broader framework of such classes for p -arithmetic groups which will give a conceptual geometric framework for understanding p -adic limiting constructions in arithmetic in general, including for example the theory of Stark–Heegner points and p -adic analogues

of the Kudla program on modular generating series of arithmetic cycles (see, e.g., [DPV1], [DGL23]).

In this paper, we realize this modular approach to explicit class field theory in the setting of rigid analytic Eisenstein classes for SL_n .

We explain our main result in more detail. Let $\mathcal{X}_p := \mathbb{P}^{n-1}(\mathbb{C}_p) - \cup_{\alpha} H_{\alpha}$ be Drinfeld's symmetric domain, where α runs over rational $\mathbb{Q}_p$ -hyperplanes H_{α} . It has a natural structure of rigid analytic space, and we denote by $\mathcal{A} = \mathcal{O}_{\mathcal{X}_p}$ its ring of rigid analytic functions. The group $\mathrm{GL}_n(\mathbb{Q}_p)$ acts on $\mathcal{X}_p$ and on $\mathcal{A}$ via left matrix multiplication. Let $f \in \mathbb{Z}_{\geq 1}$ be a positive integer such that $p \equiv 1 \pmod{f}$, and let c be an auxiliary prime with $c > n + 1$. Attached to this data, we construct a rigid analytic class

$$J_f \in H^{n-1}(\Gamma_1^p(f), \mathcal{A}^{\times}/p^{\mathbb{Z}}),$$

where $\Gamma_1^p(f)$ is the stabilizer of the vector e_1/f inside $\mathrm{SL}_n(\mathbb{Z}[1/p])$. We view this class as a generalization of the Siegel unit $c_{g(1/f,0)} \in H^0(\Gamma_1(f), \mathcal{O}_{\mathcal{H}}^{\times})$ introduced above.

Parallel to the case of Siegel units, J_f can be evaluated at special points on $\mathcal{X}_p$ attached to totally real fields. Let F be a totally real field with ring of integers $\mathcal{O}$ such that p is inert in F . Let $\mathfrak{a}$ be a fractional ideal of F prime to pf and consider $\tau \in F^n$ a positively oriented $\mathbb{Z}$ -basis of $\mathfrak{a}^{-1}$ such that $\tau^t \cdot e_1$ is a unit modulo f . Since p is inert, it follows that $[\tau] \in \mathcal{X}_p$. The point $[\tau]$ in $\mathcal{X}_p$ is special in the sense that its stabilizer in $\mathrm{SL}_n(\mathbb{Q}_p)$ contains the norm one elements in F . In particular, such stabilizer contains the subgroup $U_F(f) \subset \Gamma_1^p(f)$ isomorphic to the totally positive global units in F congruent to 1 modulo f . Since this group is isomorphic to $\mathbb{Z}^{n-1}$, we can consider $c_{U_F(f)} \in H_{n-1}(U_F(f), \mathbb{Z})$ a fundamental class. Then the value of J_f at τ is defined as

$$J_f[\tau] := J_f(c_{U_F(f)})(\tau) \in \mathbb{C}_p^{\times}/p^{\mathbb{Z}},$$

where the right hand side can be interpreted as evaluating a representative of J_f at the fundamental class $c_{U_F(f)}$, which yields a well-defined element in the coinvariants $(\mathcal{A}^{\times}/p^{\mathbb{Z}})_{U_F(f)}$, and then evaluate the corresponding function at $[\tau]$.

To describe the arithmetic meaning of this value, let $H = H_{\mathfrak{f}}$ be the narrow Hilbert class field of F of modulus $\mathfrak{f} := f\mathcal{O}$. The embedding $\bar{\mathbb{Q}} \subset \bar{\mathbb{Q}}_p$ determines a prime $\mathfrak{P}$ of H above $p\mathcal{O}_F$. Denote by $\mathcal{O}_H[1/p]_{-}^{\times}$ the subgroup of p -units of H where every complex conjugation of H acts by -1 . Attached to $\mathfrak{P}$ and c , the proof of the Brumer–Stark conjecture in [DK23] and [Das+23] implies that there is a Brumer–Stark unit $u \in \mathcal{O}_H[1/p]_{-}^{\times}$ whose valuations at primes above p are related to c -smoothed partial L -functions of the extension H/F .

Theorem 1.1 (Theorem 6.10). *Consider the same notation as above. The value of the rigid analytic class J_f at the special point $[\tau] \in \mathcal{X}_p$ is an algebraic number in an abelian extension of F . More precisely, we have the equality*

$$J_f[\tau] = u^{\sigma_{\mathfrak{a}}} \pmod{p^{\mathbb{Z}} \cdot \mu(F_p^{\times})},$$

where $u \in \mathcal{O}_{H_{\mathfrak{f}}}[1/p]_{-}^{\times}$ is the Brumer–Stark unit introduced above and $\sigma_{\mathfrak{a}} \in \mathrm{Gal}(H_{\mathfrak{f}}/F)$ is the Frobenius attached to $\mathfrak{a}$.

Remark 1.2. The congruence condition on p is to ensure full splitting of $p\mathcal{O}_F$ in $H_{\mathfrak{f}}$; we expect the same methods to yield Brumer–Stark unit specializations in arbitrary abelian extensions of F in which p fully splits. We restrict ourselves to the ray class setting for simplicity, and because this already suffices for the purposes of explicit class field theory of the maximal abelian extension of F [DK23].

In fact, our rigid cocycles for $\Gamma_1^p(f)$, for various primes p and levels f , all come from specializing a single object

$$H^{n-1}(\mathrm{SL}_n(\mathbb{Z}_{(c)}), \mathbf{D})$$

where $\mathbf{D}$ is a module of adelic distributions, and c is an auxiliary prime.

Remark 1.3. We remark that actually believe that extensions of our methods may be used to show that this cocycle specializes to all Brumer–Stark units for all primes attached to any prime p different from c , for all totally real fields of degree n , *regardless* of the splitting type of p in F : however, for non-inert p , these formulas cannot be literally interpreted in terms of standard evaluations of rigid analytic cocycles. Their form is very suggestive of a kind of “degenerate evaluation at the boundary of the p -adic symmetric space” of these cocycles, but we are not currently able to make this precise.

Our proof that these specializations yield unit formulas, as in the case of Dasgupta and collaborators’ formulas, relies on the proof of the integral Gross–Stark conjecture from the work of Dasgupta–Kakde.

Our classes therefore continue the program of establishing the arithmetic significance of rigid analytic classes, showing they give a “ p -modular” or “ p -automorphic” way to systematize the solution to Hilbert’s twelfth problem for totally real fields into a single analytic object, uniform across all fields of a given degree, just as the j -function or Siegel units do in the imaginary quadratic setting. The explicit nature of our cocycles also leads to simplified formulas for these Brumer–Stark units, which may be of computational interest.

1.2. Outline of article. Our article is structured as follows: in Section 2, we construct our distribution-valued cocycles

$${}^c\Theta_{\mathbf{D}}^{(n)} \in H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}_{(c)}), \mathbf{D}(\mathbb{X}, \mathbb{Z})^{(0)}/\delta_0).$$

where c is an auxiliary prime, and the coefficients are a certain module of distributions modulo atoms at zero.

We obtain this class by applying Cartier duality to the coefficients a cohomology class valued in functions $\mathbf{T}$ (or rather the canonically trivial line bundle $\Omega_{\mathbf{T}}^n$) holomorphic on all prime-to- c torsion sections. The method for constructing this latter class is taken from the second author’s previous article [X1]: roughly speaking, it is given by a homogeneous $(n-1)$ -cocycle of the form

$$(\gamma_1 : \dots : \gamma_n) \mapsto \left((\gamma_1)_1 \ \dots \ (\gamma_n)_1 \right)_* d\log(z_1 - 1) \wedge \dots d\log(z_n - 1)$$

(where $(\gamma_i)_1$ denotes the first column of γ_i) but shifted by “stabilization” data coming from c -torsion points ensuring holomorphicity at all prime-to- c torsion (including in special fibers over

places different from c). This construction comes from a nearly-exact complex constructed from rational cones, which parameterizes certain Dolbeault currents on an algebraic torus $(\mathbb{C}^\times)^n$: see Theorem 2.5 and Proposition 2.10 for details.

In Section 3, we show, using general homological algebra, that the differential-form valued cohomology class comes from an *equivariant Eisenstein class* defined in analytic coherent cohomology on $(\mathbb{C}^\times)^n$, defined solely via its residues and a trace-fixed condition. These methods closely resemble, and were heavily inspired by, the application of equivariant coherent cohomology of Kings–Sprang’s treatment of p -adic L -functions of algebraic Hecke characters [KS].

This enables us to compute the specializations of ${}^c\Theta_{\mathbf{D}}^{(n)}$ via analytic representatives for these equivariant Eisenstein classes in Section 4, which we construct following a mixture of the techniques used to find analogous representatives by Kings–Sprang [KS] and Bergeron–Charollois–García in [BCG20] and [BCG23]. The periods of these representatives along non-split tori we calculate to be L -values attached to abelian extensions of the corresponding totally real fields (Theorem 4.15).

In Section 5, combining these L -value period computations and the (known) integral Gross–Stark conjecture [DK23], we apply the “patching” strategy of Dasgupta [Das08] to prove (in Section 5.5) that certain specializations of ${}^c\Theta_{\mathbf{D}}^{(n)}$ yield algebraic Brumer–Stark units. Finally, we pass from distribution-valued cocycles to $\mathcal{A}^\times$ -valued cocycles using a multiplicative “Van der Put” lift in Section 6, reinterpreting our results in terms of values at special moduli of rigid analytic cocycles.

1.3. Eisenstein cocycles in the literature. As mentioned, our main construction heavily uses the “toric Eisenstein cocycles” constructed by the second author on an algebraic torus in [X1]. This article is also similar in spirit to our previous work [RX1], where we produced an Eisenstein class in

$$H^{n-1}(\mathrm{SL}_n(\mathbb{Z}), \mathcal{A}_{\mathcal{L}}),$$

using the cohomology of a *topological* n -torus, where $\mathcal{A}_{\mathcal{L}}$ denotes the log-rigid analytic functions on $\mathcal{X}_p$, i.e. functions which are the composition of invertible rigid analytic functions with the Iwasawa logarithm. In loc. cit., we showed its specializations are closely related to p -adic logarithms of Gross–Stark units. The approach of the present article can be viewed as an explicit $\mathbb{Z}$ -valued refinement of loc. cit. to the p -arithmetic setting.

However, we do not here actually prove our two cocycles coincide (after appropriate forgetful maps and restrictions). Morally, the relationship between the two is the relationship between an equivariant polylogarithm on an *algebraic* torus versus a *topological* torus, which corresponds to a reflective functional equation between weights $s = 0$ and 1 .

There are many other antecedent “Eisenstein cocycle” constructions in the literature which are quite relevant, but beyond those mentioned for other reasons above, we do not review them in detail here: such a review would have huge overlap with the reviews in [X1] and [RX1], which the interested reader can consult. We note among these [BKL] and [CDG]; the latter “Shintani” approach especially is closely related to the second author’s approach pioneered in [X1] used in the present article, as described in the latter half of loc. cit.

We also wish to especially highlight the “adelic” distribution-valued Eisenstein classes of Galanakis–Spiess [GS] is also an S -arithmetic cocycle with coefficients in integral distributions very similar to those we obtain. Their approach manages to construct these classes in a more purely formal algebraic way, by doing sheaf cohomology in the pertinent profinite settings (specifically, a “solenoidal” inverse limit of tori along multiplication maps) using a carefully selected Grothendieck topology. By comparison, our symbol-based construction uses only explicit and elementary techniques. We expect our distribution-valued classes essentially coincide with theirs, which should in particular imply the expected relationship with [RX1] noted above, as Galanakis–Spiess’ construction may be projected down to the equivariant polylogarithm construction for a topological torus of loc. cit.

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1.5. AI use statement, in conformance with the Leiden Declaration: We used LLM assistance, though only during the last stages of this project after a majority of writing was complete. We used primarily Claude Opus and ChatGPT Astra for proofreading, checking, literature searches, and general consultation. Most notably, Claude Opus suggested the correct form of the analytic coherent representatives in Section 4.

2. DISTRIBUTION-VALUED COCYCLES

In this section, we construct an Eisenstein class with coefficients in $\mathbb{Z}$ -valued adelic distributions. More precisely, let $c > n + 1$ be an auxiliary prime, $\mathbb{X} := \prod_{(\ell, c)=1} \mathbb{Z}_\ell^n$, $\mathbb{Y} := \mathbb{X} \otimes \mathbb{Q}$, and consider $\mathbf{D}(\mathbb{Y}, \mathbb{Z})^{(0)}$ the space of $\mathbb{Z}$ -valued measures on $\mathbb{Y}$ that are invariant under multiplication by integers a prime to c . We construct a class

$${}_c\Theta^{\mathbf{D}} \in H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}_{(c)}), \mathbf{D}(\mathbb{Y}, \mathbb{Z})^{(0)} / \mathbb{Z}\delta_0),$$

where δ_0 denotes the dirac measure at $0 \in \mathbb{Y}$, and explain a procedure to find representatives of it. Our construction follows the strategy of the second-named author in [X1] used to parametrize toric polylogarithm elements using the geometry of S^{n-1} , but it is technically lighter as we only need the differential forms version of this construction. It can be summarized as follows:

Rational polyhedral decompositions of S^{n-1} give an augmented complex of spherical chains which admits an action of $\mathrm{GL}_n^+(\mathbb{Q})$ and whose only homology is generated by the fundamental class in degree $n - 1$. After quotienting by this class, the complex becomes exact. Successively lifting an invariant element through this exact complex produces a group cohomology class of degree $n - 1$, with coefficients in top-dimensional chains modulo the fundamental class.

To give arithmetic meaning to this construction, we consider a realization map from spherical chains to currents on the torus $\mathbf{T} := \mathbb{G}_m^n$ with singularities along hyperplanes. This realization map is constructed using Shintani-style cone generating series.

We then consider a c -stabilized version of this construction to control the polar divisors of the image of the realization map, ensuring that it consists of forms that are regular on every prime to c torsion group scheme over $\mathbb{Z}[1/c]$. By restricting these forms to prime to c torsion

subgroups, and dividing by $\omega = d\log z_1 \wedge \cdots \wedge d\log z_n$, we obtain a compatible system of functions. In the spirit of Katz [Katz], a version of Cartier duality identifies this compatible system of functions with our desired distribution on $\mathbb{X}$, which is viewed as the space of finite order characters (prime to c) on $\mathbf{T}$.

2.1. Homology of spheres.

Definition 2.1. A *rational polyhedral decomposition* of S^{n-1} is the set of intersections $\sigma \cap S^{n-1}$ for $\sigma \in \Sigma$, where Σ is a complete rational polyhedral fan in $\mathbb{R}^n$. That is, Σ is a finite collection of subsets of $\mathbb{R}^n$ satisfying:

- (1) Every $\sigma \in \Sigma$ is generated by a finite set of vectors in $\mathbb{Q}^n$, i.e.

$$\sigma = \left\{ \sum_{j=1}^k \lambda_j w_j : \lambda_j \geq 0 \right\} \quad \text{for some } w_j \in \mathbb{Q}^n.$$

- (2) Every $\sigma \in \Sigma$ is strongly convex ($\sigma \cap -\sigma = \{0\}$).

- (3) If $\sigma \in \Sigma$ and τ is a face of σ , then $\tau \in \Sigma$.

- (4) If $\sigma, \tau \in \Sigma$, their intersection $\sigma \cap \tau$ is a face of both σ and τ .

- (5) The union of all cones covers the space: $\bigcup_{\sigma \in \Sigma} \sigma = \mathbb{R}^n$.

We will sometimes abuse notation and denote the rational polyhedral decomposition of S^{n-1} simply by Σ .

Denote by $\mathcal{P}$ the set of rational polyhedral decompositions of S^{n-1} . Given two rational polyhedral decompositions of S^{n-1} , denoted Σ_1 and Σ_2 , we say that Σ_2 refines Σ_1 , and write $\Sigma_1 \subset \Sigma_2$, if every cone of Σ_2 is contained inside Σ_1 . Given two rational polyhedral decompositions Σ_1 and Σ_2 , there exists a rational polyhedral decomposition Σ' such that $\Sigma_1 \leq \Sigma'$ and $\Sigma_2 \leq \Sigma'$, simply by superimposing all the faces of both decompositions, and taking the resultant chambers. We have the following standard triangulation result; e.g. see [CLS, Proposition 11.1.7].

Lemma 2.2. *Any rational polyhedral decomposition can be refined to a rational simplicial decomposition of S^{n-1} , i.e. a decomposition whose cones are all simplicial cones (i.e. whose intersections with S^{n-1} are simplices).*

For any given rational polyhedral decomposition Σ , we can consider the corresponding augmented homology complex

$$(C_\Sigma)_\bullet : (C_\Sigma)_{n-1} \longrightarrow (C_\Sigma)_{n-2} \longrightarrow \cdots \longrightarrow (C_\Sigma)_0 \longrightarrow (C_\Sigma)_{-1} := \mathbb{Z}.$$

The homology of this complex computes reduced homology of S^{n-1} . Finally, we define the complex $C_\bullet = (C_i)_i$ by taking a direct limit over refinements of rational polyhedral decompositions. Namely for every i

$$C_i := \varinjlim_{\Sigma \in \mathcal{P}} (C_\Sigma)_i.$$

The homology of $C_\bullet$ computes the reduced homology of S^{n-1} .

We will also be interested in certain subcomplexes of $C_\bullet$. Let $c \in \mathbb{Z}_{\geq 0}$ be an auxiliary prime number. Consider $\lambda \in \mathbb{P}^{n-1}(\mathbb{Z}/c\mathbb{Z})$.

Definition 2.3. A rational polyhedral decomposition Σ of S^{n-1} is λ avoiding if for every wall, which corresponds to a unique rational hyperplane H with primitive normal vector $v_H \in \mathbb{Z}^n$, we have that $v_H \cdot \lambda \not\equiv 0 \pmod{c}$.

Note that as with the unrestricted case, it follows simply from superimposition that given two rational polyhedral decompositions Σ_1 and Σ_2 which are λ -avoiding, there exists Σ' which is λ -avoiding and satisfies $\Sigma_1 \leq \Sigma'$, $\Sigma_2 \leq \Sigma'$.

Lemma 2.4. *The set of λ -avoiding polyhedral decompositions is non-empty.*

Proof. For example, we may fix a single mod- c hyperplane avoiding λ ; by the Chinese remainder theorem, the set of lifts to hyperplanes in $\mathbb{Q}^n$ is dense, and we may simply slice along sufficiently many such hyperplanes that all resulting polyhedral chambers are strongly convex. $\square$

As a consequence, we can define ${}^\lambda C_\bullet = ({}^\lambda C_i)_i$ as

$${}^\lambda C_\bullet := \varinjlim_{\Delta \text{ is } \lambda \text{ avoiding}} ({}^\lambda C_\Delta)_i$$

and view it as a subcomplex of $C_\bullet$. Moreover, the homology of ${}^\lambda C_\bullet$ calculates the reduced homology of S^{n-1} .

2.2. Realization map to currents on $(\mathbb{C}^\times)^n$. Given a d -dimensional complex manifold M , write $\mathcal{D}_M^{p,q}$ for the smooth (p, q) -currents on M , i.e. the topological dual

$$\mathcal{D}_M^{p,q} := \text{Hom}_c(A_{M,c}^{n-p,n-q}, \mathbb{C}),$$

where $A_{M,c}^{n-p,n-q}$ denotes the compactly supported smooth forms on M of holomorphic degree p , respectively anti-holomorphic degree q . The antiholomorphic derivative $\bar{\partial}: \mathcal{D}_M^{p,q} \rightarrow \mathcal{D}_M^{p,q+1}$ is defined by

$$(\bar{\partial}c)(\eta) := (-1)^{\deg c} c(\bar{\partial}\eta).$$

With this differential, the $(p, \bullet)$ -currents on M form the distributional Dolbeault complex $\mathcal{D}_M^{p,\bullet}$ of holomorphic degree p , resolving the sheaf of holomorphic p -differentials Ω_M^p . We refer to [KS, Appendix A] and [KS, §2] for additional properties of this construction: pushforward, the localization sequence, purity, etc.

Let $\mathbf{T} := \mathbb{G}_m^n$ with its left $\text{GL}_n(\mathbb{Z})$ -action; we will write $X(\mathbf{T})$ (or usually just X , since we are always working with a fixed torus) for its character lattice $\text{hom}(\mathbf{T}, \mathbb{G}_m^n)$, which we consider as a left $\text{GL}_n(\mathbb{Z})$ -module via pullback by the inverse.

In the rest of the section, we construct a map of complexes

$$\Psi: C_{\bullet-1} \longrightarrow \mathcal{D}_{\mathbf{T}}^{n,n-\bullet}.$$

More precisely, we wish to prove the following result.

Theorem 2.5. *The assignment defined by*

$$C_{k-1} \longrightarrow (\mathcal{D}_{\mathbf{T}}^{n,n-k})^{(0)}$$

$$[\Delta(r_1, \dots, r_k)] \mapsto (2\pi i)^{n-k} \begin{pmatrix} \tilde{r}_1 & \dots & \tilde{r}_k \end{pmatrix}_* \left[\left(\prod_{i=1}^k \frac{z_i}{z_i - 1} \right) d \log z_1 \wedge \dots \wedge d \log z_k \right]$$

in degree $k \geq 1$, and by $1 \in C(n)_{-1} \mapsto (2\pi i)^n \delta_1 \in \mathcal{D}_{\mathbf{T}}^{n,n}$ in degree -1 , is a well-defined $\mathrm{GL}_n^+(\mathbb{Q})$ -equivariant morphism of complexes

$$\Psi = (\Psi_{\bullet-1}): C(n)_{\bullet-1} \longrightarrow (\mathcal{D}_{\mathbf{T}}^{n,n-\bullet})^{(0)}$$

sending the fundamental class $[S^{n-1}]$ to the determinant form $\omega := d \log z_1 \wedge \dots \wedge d \log z_n$.

Here, $\Delta(r_1, \dots, r_k)$ denotes a simplex, and we are using that every rational polyhedral decomposition can be refined to a rational triangulation to describe this map. Moreover, each $\tilde{r}_i$ is an integer vector on the line generated by r_i . Finally, the decoration (0) on the target means the part fixed by the scalar pushforwards $[a]_*$ for every $a \in \mathbb{N}$.

We will prove this theorem in several steps. Most of them are focused on the study of this assignment in degree $n - 1$. The general case will follow from this one.

Proposition 2.6. *The assignment Ψ is well-defined in degree $n - 1$. Namely,*

$$\Psi_{n-1}: C_{n-1} \longrightarrow \mathcal{D}_{\mathbf{T}}^{n,0}$$

described in the previous theorem is a well-defined function.

Proof. To prove this proposition, we will express Ψ_n as a composition of three maps that are well-defined. Let $\mathcal{K}$ be the module of $\mathbb{Z}$ -valued functions on $\mathbb{R}^n$ generated by indicator functions of closed convex rational polyhedral cones, with left $\mathrm{GL}_n(\mathbb{Q})$ -action given by push-forward. Let $\mathcal{K}_0 \subset \mathcal{K}$ be the submodule consisting of functions that are 0 on a dense open set. For each rational polyhedral decomposition Σ of S^{n-1} , we have a map

$$G_{\Sigma}: C_{\Sigma, n-1} \longrightarrow \mathcal{K}/\mathcal{K}_0, \quad [\Sigma] \longmapsto [1_{\Sigma}].$$

Since $\mathcal{K}_0$ contains the characteristic functions of the faces of every cone, the collections of maps $(G_{\Sigma})_{\Sigma}$ induce a map

$$G := (G_{\Sigma})_{\Sigma}: C_{n-1} \longrightarrow \mathcal{K}/\mathcal{K}_0.$$

Second, given a cone $C \subset \mathbb{R}^n$, consider its dual cone

$$C^{\vee} := \{w \in \mathbb{R}^n \mid \langle w, v \rangle \geq 0 \text{ for all } v \in C\}.$$

From convexity theory (cf. [Bv, Chapter IV, §1]), there is a linear involution coming from *conical duality*, $D: \mathcal{K} \rightarrow \mathcal{K}$, defined as $D(1_C) = 1_{C^{\vee}}$. Thus, if we define $\mathcal{L} := D(\mathcal{K}_0)$, then $\mathcal{L}$ may be described as generated by indicator functions of cones containing a positive dimensional subspace: this is because the dual of a polyhedral cone of codimension- d is a cone containing a dimension- d subspace, and vice versa. Duality descends therefore to an isomorphism

$$D: \mathcal{K}/\mathcal{K}_0 \xrightarrow{\sim} \mathcal{K}/\mathcal{L}.$$

Moreover, for every $\gamma \in \mathrm{GL}_n(\mathbb{Q})$, $D \circ \gamma = \gamma^{-T} \circ D$.

Finally, consider the “cone generating function” map

$$\Phi: \mathcal{K}/\mathcal{L} \longrightarrow \mathbb{Z}((z_1, \dots, z_n)), \quad 1_C \longmapsto \sum_{v \in C \cap \mathbb{Z}^n} z^{-v},$$

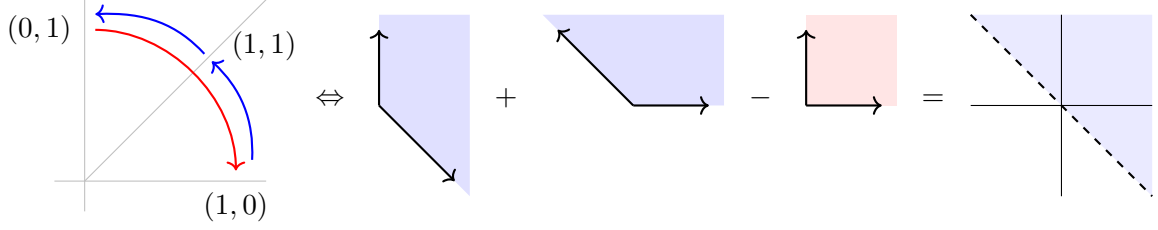


FIGURE 1. The spherical chains identity

$$\Delta(e_1, e_1 + e_2) + \Delta(e_1 + e_2, e_2) - \Delta(e_1, e_2) = 0,$$

and its corresponding dual cone identity. The generating functions of the three cones sum to the conical generating function of the half-plane “wedge”, which is zero. Note this half-plane is precisely the dual of the cone $\mathbb{R}_{\geq 0}(1, 1)$, corresponding to the fact that we would have a $(1, 1)$ “left over” in the chain identity, if spherical chains kept track of their lower-dimensional data.

Observe that Φ is valued in rational functions on $\mathbf{T}$ with poles along divisors of the form $\{\chi = 1\}$ for various characters χ of $\mathbf{T}$ [CDG, Proposition 2.1].

We now claim that the composition

$$\tilde{\Phi}: C(n)_{n-1} \xrightarrow{G} \mathcal{K}/\mathcal{K}_0 \xrightarrow{D} \mathcal{K}/\mathcal{L} \xrightarrow{\Phi \cdot \omega} \mathcal{D}_{\mathbf{T}}^{n, n-k}$$

is equal to Ψ_n , which will imply that Ψ_n is well-defined. Indeed, it is a computation to verify that

$$\tilde{\Phi}(\Delta(e_1, \dots, e_n)) = \Phi(\Delta(e_1, \dots, e_n))\omega = \Psi(\Delta(e_1, \dots, e_n)).$$

Now, for every simplex $\Delta(r_1, \dots, r_n)$, let

$$R = (r_1 \dots r_n) \in \mathrm{GL}_n(\mathbb{Q}), \quad \tilde{R} = (\tilde{r}_1 \dots \tilde{r}_n) \in M_n(\mathbb{Z}) \cap \mathrm{GL}_n(\mathbb{Q})$$

be such that $\tilde{r}_i$ is proportional to r_i . Note that $\tilde{R}C(e_1, \dots, e_n) = C(r_1, \dots, r_n)$. Then,

$$\begin{aligned} \tilde{\Phi}(\Delta(r_1, \dots, r_n)) &= \Phi(1_{C(r_1, \dots, r_n)^\vee})\omega \\ &= \Phi(1_{\tilde{R}^{-T}C(e_1, \dots, e_n)^\vee})\omega \\ &= \frac{1}{|\det \tilde{R}|} \mathrm{Tr}_{\tilde{R}} \Phi(1_{C(e_1, \dots, e_n)})\omega \\ &= \tilde{R}_* (\Phi(1_{C(e_1, \dots, e_n)})\omega) \\ &= \tilde{R}_* \Psi(\Delta(e_1, \dots, e_n)) = \Psi(\Delta(r_1, \dots, r_n)). \end{aligned}$$

as desired. □

Corollary 2.7. *The assignment Ψ is well-defined at every degree.*

Proof. Let us decorate with superscript (k) our constructions for a k -dimensional torus. To show the map $\Psi_k^{(n)}: C_{k-1}^{(n)} \rightarrow (\mathcal{D}_{\mathbf{T}^{(n)}}^{n, n-k})^{(0)}$ is well-defined, note that all subdivision relations occur along individual $(k-1)$ -great circles. Thus, it suffices to show well-definedness along some map $S^{k-1} \rightarrow S^{n-1}$ and its corresponding inclusion $M_*: C_{k-1}^{(k)} \rightarrow C_{k-1}^{(n)}$, which we view as

given by some matrix M . But by pushforward functoriality, this restriction of $\Psi_k^{(n)}$ is simply the composite

$$C_{k-1}^{(k)} \xrightarrow{\Psi_{k-1}^{(k)}} \mathcal{D}_{\mathbf{T}^{(k)}}^{k,0} \xrightarrow{M_*} \mathcal{D}_{\mathbf{T}^{(n)}}^{n,n-k}$$

which is the top-dimensional case of the previous proposition. $\square$

Proposition 2.8. *The formula for Ψ defines a map of complexes.*

Proof. Using that Ψ is $\mathrm{GL}_n(\mathbb{Q})$ -equivariant, it is enough to see that

$$\Psi(\partial\Delta(e_1, \dots, e_k)) = \bar{\partial}\Psi(\Delta(e_1, \dots, e_k)).$$

By the same argument as the previous corollary, using functoriality of pushforward, we may assume $k = n$. Then since the only poles of our standard meromorphic form

$$\Psi(\Delta(e_1, \dots, e_n)) = d\log(z_1 - 1) \wedge \dots \wedge d\log(z_n - 1)$$

are log poles along the hyperplanes $z_i = 1$, the Poincaré-Lelong formula exactly says that the distributional derivative is supported on these hyperplanes, with the term supported on the i th one being precisely being

$$(-1)^{i-1} d\log(z_1 - 1) \wedge \dots \wedge d\log(\widehat{z_i - 1}) \wedge \dots \wedge d\log(z_n - 1) \delta_{z_i=1}$$

whose sum over i is precisely

$$\Psi\left(\sum_{i=1}^n (-1)^{i-1} \Delta(e_1, \dots, \widehat{e_i}, \dots, e_n)\right)$$

as desired. $\square$

2.3. λ -smoothing. As before, let $\lambda \in \mathbb{P}^{n-1}(\mathbb{Z}/c)$ be a c -torsion line; under the uniformization $\mathbf{T} \cong (\mathbb{C}/\mathbb{Z})^n$ (equivariant for all choices of action we have made so far), we may view it as a subgroup of c -roots of unity inside our algebraic torus. Restricting to the λ -avoiding chains, we find:

Proposition 2.9. *The image under Ψ of ${}^\lambda C(n)_{n-1}$ in $\mathcal{D}_{\mathbf{T}}^{n,0}$ lands in the image of meromorphic n -forms on $\mathbf{T}$ whose polar divisors avoid $\lambda - \{1\} \subset \mu_c^n$ (over the generic fiber).*

Proof. Let $Z \in {}^\lambda C_{n-1}$. The polar divisors of the distribution $\Psi(Z)$ are given are precisely the support of $\bar{\partial}\Psi(Z)$, as the vanishing of $\bar{\partial}$ outside this support implies holomorphicity of $\Psi(Z)$ by the first part of the Dolbeault resolution.

Using that Ψ is a map of complexes, $\bar{\partial}\Psi(Z) = \Psi(\partial Z)$. Since $Z \in {}^\lambda C_{n-1}$, ∂Z is contained in a union rational hyperplanes that do not contain the line λ modulo c . In particular, ∂Z can be triangulated with simplices all contained in (multiplicative) hyperplanes that do not contain the line λ modulo c . But finally, if $\Delta \in C_{n-2}$ is such a simplex, we see from the expression of $\Psi(\Delta)$ that its support will be contained in one of these hyperplanes, and therefore, it will be disjoint from λ . $\square$

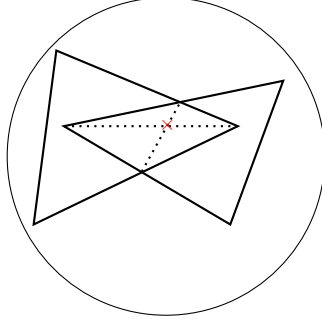


FIGURE 2. The most naive triangulations of the superimposed pair of triangles on S^2 fail to be λ -avoiding, because the red $\times$ marks a point whose corresponding ray reduces mod c to λ . One may simply instead consider the quadrilateral itself as a homology cycle on S^2 , whose formal boundary is equal to a sum of four 1-simplices: the pictured triangulation is valid in $C(n)_{n-1}$, but not in ${}^\lambda C(n)_{n-1}$. As remarked below, we expect that triangulating everything *is* in fact always possible even in this restricted setting, but have no need of it.

In light of the preceding corollary, we define the $\mathrm{GL}_n(\mathbb{Z}_{(c)})$ -equivariant c -stabilized realization map

$${}^c\Psi : \bigoplus_{\lambda \in \mathbb{P}^{n-1}(\mathbb{Z}/c)} {}^\lambda C(n)_\bullet \longrightarrow (\mathcal{D}_{\mathbf{T}}^{n,n-\bullet-1})^{(0_c)}, \quad (Z)_\lambda \longmapsto \sum_{x \in \lambda \setminus \{1\}} t_x^* \Psi(Z) \quad (1)$$

where here $t_x : \mathbf{T} \rightarrow \mathbf{T}, z \mapsto xz$ is the translation-by- x map, and (0_c) means the same thing as (0) , but restricted only to scalars prime to c . Observe that the source of ${}^c\Psi$ has top homology $\mathbb{Z}^{\mathbb{P}^{n-1}(\mathbb{Z}/c)}$, coming from the fundamental classes of each of the $\mathbb{P}^{n-1}(\mathbb{Z}/c)$ -many spheres, but that each of these fundamental classes just maps to the same form $(c-1)\omega$. In other words, ${}^c\Psi_{n-1}$ actually factors through the quotient by the sum-zero subspace

$$\ker(\mathbb{Z}^{\mathbb{P}^{n-1}(\mathbb{Z}/c)} \rightarrow \mathbb{Z})$$

of the top homology; we denote this quotiented complex by ${}^c C(n)_\bullet$.

Let $\mathcal{O}_{(c)} \hookrightarrow (\mathcal{D}_{\mathbf{T}}^{n,0})^{(0_c)}$ denote the image of ${}^c\Psi_{n-1}$. It then follows from the preceding theorem and corollary:

Proposition 2.10. *The distributions in $\mathcal{O}_{(c)}$ are in the image of trace-fixed (away from c) meromorphic Kähler n -forms on (the scheme) $\mathbf{T}$ over $\mathbb{Z}[1/c]$ which are regular on all prime-to- c torsion over the entire base.*

Proof. A polar hypersurface of the image of ${}^c\Psi$ is a group subtorus S translated by a c -torsion point x which does not lie on S over $\mathbb{C}$, and therefore also over $\mathbb{Z}[1/c]$, since over this base (avoiding characteristic c) the complex c -torsion of an algebraic torus injects into the c -torsion over every fiber. Thus xS does not pass through the identity over $\mathbb{Z}[1/c]$. Moreover, for any N prime to c and exact N -torsion point y , $y \in xS$ would imply the cN -torsion point $x^{-1}y$ lay in S , and thus, taking N th powers, the line of c -torsion generated by x lay in S , which is a contradiction. $\square$

We note that we can really identify $\mathcal{O}_{(c)}$ with these meromorphic forms, because they inject into $\mathcal{D}_{\mathbf{T}}^{n,0}$: if we restrict away from the polar hyperplane locus, the distributions in $\mathcal{O}_{(c)}$ become $\bar{\partial}$ -closed, hence may be identified with holomorphic n -forms on this Zariski open.

2.4. Measures. We now see how to obtain $\mathbb{Z}$ -valued adelic distributions from the image of ${}^c\Psi_{n-1}$. Integrality at primes away from c will follow from Cartier duality for the various finite flat group schemes $\mathbf{T}[k]$ over $\mathbb{Z}[1/c]$ following the point of view written by Katz [Katz] (written there for formal groups over adic rings). We can control the denominators at c via the explicit formulas.

Given a locally profinite space Y and an abelian group A , let $\mathbf{D}(Y, A)$ denote the distributions against A -valued Schwartz (locally constant, compactly supported) functions on Y . Let $\mathbb{X}_0$ be the inverse limit over all $(k, c) = 1$ of $\varprojlim X/kX$ (i.e. the characters of $\mathbf{T}[k]$); this is a profinite space which may be identified $\mathrm{GL}_n(\mathbb{Z})$ -equivariantly with $(\hat{\mathbb{Z}}^{(c)})^n$ (with the pullback-by-inverse left action) via

$$((\hat{\mathbb{Z}}^{(c)})^n)^\vee \xrightarrow{\sim} \varprojlim X_k, \quad (L : (\hat{\mathbb{Z}}^{(c)})^n \rightarrow \hat{\mathbb{Z}}^{(c)}) \mapsto (x \in \mathbf{T}[k] \mapsto \exp(2\pi i L(t)))$$

where $t \in (k^{-1}\mathbb{Z}/\mathbb{Z})^n$ uniformizes $x \in \mathbf{T}[k]$ and L is viewed as a function on $(\hat{\mathbb{Q}}^{(c)})^n$ via linear extension.

Then we have the following Cartier duality isomorphism, just as in [Katz, Theorem 1]:

Proposition 2.11. *There is an $(M_n(\mathbb{Z}) \cap \mathrm{GL}_n(\mathbb{Z}_{(c)}))$ -equivariant isomorphism*

$$\varphi : \varprojlim_{(k,c)=1} \mathcal{O}(\mathbf{T}[k]) \xrightarrow{\sim} \mathbf{D}(\mathbb{X}_0, \mathbb{Z}[1/c]), \quad f \mapsto \mu_f$$

characterized by

$$\int_{\mathbb{X}_0} \chi(x) d\mu_f(\chi) = f(x),$$

where the action on the source and target are both by pushforward.

Moreover, the image of the function $1 \in \mathcal{O}(G)$ is the Dirac measure supported on the trivial character $1 \in \mathbf{T}[k]$ for every k , that we denote by δ_0 .

We have a $\mathrm{GL}_n^+(\mathbb{Z}_{(c)})$ -equivariant map from contracting with the determinant vector field and restriction to G

$$\mathrm{restrict} : \mathcal{O}_{(c)} \longrightarrow \varprojlim \mathcal{O}(\mathbf{T}[k])^{(0)}(-\det), \quad \eta \mapsto \frac{\eta}{\omega}|_G$$

which lands more specifically in holomorphic functions on $\mathbf{T}$ which transform with weight n under pushforward $[a]_*$ for any $(a, c) = 1$, in the sense that $[a]_* f = a^n f$. Here, we use Proposition 2.10 to see that the functions in $\mathcal{O}_{(c)}$ indeed are regular over all the k -torsion over the whole base $\mathbb{Z}[1/c]$.

Then the characterizing property in the preceding proposition then implies that the corresponding distributions lie in the submodule

$$\mathbf{D}(\mathbb{X}_0, \mathbb{Z}[1/c])^{(0)} \subset \mathbf{D}(\mathbb{X}_0, \mathbb{Z}[1/c])$$

of distributions invariant under all $[a]_*$ (with $([a]_*\mu)(U) := \mu(aU)$). If we write $\mathbb{X} := \mathbb{X}_0 \otimes \mathbb{Q}$, we may identify

$$\mathbf{D}(\mathbb{X}_0, \mathbb{Z}[1/c])^{(0)} \cong \mathbf{D}(\mathbb{X}, \mathbb{Z}[1/c])^{(0)}$$

by extending by this scalar invariance. (Note in particular that the invariance forces the measure of $\mathbb{X}_0 - a\mathbb{X}_0$ to be zero for any $(a, c) = 1$.)

We thus may form the composite map

$${}^cC(n)_{n-1} \xrightarrow{{}^c\Psi} \mathcal{O}_{(c)} \xrightarrow{\text{restrict}} \varprojlim \mathcal{O}(\mathbf{T}[k])^{(0)}(-\det) \xrightarrow{\varphi} \mathbf{D}(\mathbb{X}_0, \mathbb{Z}[1/c])^{(0)} \cong \mathbf{D}(\mathbb{X}, \mathbb{Z}[1/c])^{(0)} \quad (2)$$

which is $\mathrm{GL}_n^+(\mathbb{Z}_{(c)})$ -equivariant.

As promised, we may also explicitly control the c -integrality of this map.

Proposition 2.12. *The image of (2) is valued in distributions with valuation at c at least $n/(1-c)$. In particular, if $c > n+1$, then it is valued in $\mathbb{Z}$.*

Proof. The values of μ_f are c -integral linear combinations of sums of terms whose denominators are of the form

$$(1 - \zeta_1) \dots (1 - \zeta_n)$$

for some n -tuple of nontrivial roots of unity $\zeta_1, \dots, \zeta_n$. (Note that the specialization at a torsion section x of a function M_*f is the same as the sum of specializations of f at the torsion sections in $M^{-1}x$.) The valuation at c of this denominator is therefore at most $n/(c-1)$, from which the result immediately follows. $\square$

Corollary 2.13. *Suppose $c > n+1$, we have a map*

$${}^cC(n)_{n-1}/H_{n-1}({}^cC(n)_\bullet) \longrightarrow \mathbf{D}(\mathbb{X}, \mathbb{Z})^{(0)}/\delta_0.$$

2.5. Construction of the cocycle valued on measures. We finally use the following lemma to construct group cocycles.

Lemma 2.14. *If a group G acts on an exact complex*

$$C_\bullet : 0 \longrightarrow C_n \longrightarrow C_{n-1} \longrightarrow \dots \longrightarrow C_0 \longrightarrow 0$$

then we have a natural map on cohomology

$$C_0^G \longrightarrow H^{n-1}(G, C_n)$$

inhomogenous cocycle representatives of which can be constructed as follows: associated to $e \in C_0^G$, pick a lift c_1 of e to C_1 , and consider the 1-cochain

$$\gamma \mapsto (\gamma - 1)c_1 \in C^1(G, C_1).$$

By exactness, this is the boundary of an element $c_2 \in C^1(G, C_2)$; we take the chain coboundary $\partial c_2 \in C^2(G, C_2)$ which again lifts to $c_3 \in C^2(G, C_3)$, etc. The lift $c_n \in C^{n-1}(G, C_n)$ is a cocycle representing the image of e in $H^{n-1}(G, C_n)$.

Proof. See [X1, Lemma 2.5]. $\square$

We can apply the previous lemma to the exact complex

$$0 \longrightarrow {}^cC(n)_{n-1}/H_{n-1} \longrightarrow {}^cC(n)_{n-2} \longrightarrow \dots \longrightarrow {}^cC(n)_0 \longrightarrow \bigoplus_{\lambda \in \mathbb{P}^{n-1}(\mathbb{Z}/c)} \mathbb{Z} \longrightarrow 0$$

and the element

$$e := (1_\lambda)_\lambda \in \bigoplus_{\lambda \in \mathbb{P}^{n-1}(\mathbb{Z}/c)} \mathbb{Z},$$

to obtain a collection of cochains

$$c_i \in C^{i-1}(\mathrm{GL}_n^+(\mathbb{Z}_{(c)}), C_i), \text{ for } i = 1, \dots, n$$

and in particular, if we view c_{n-1} with coefficients in ${}^c C(n)_{n-1}/H_{n-1}$ we obtain an $n-1$ cocycle (i.e. it is closed). We denote the corresponding class by

$${}^c \Theta^{S^{n-1}} \in H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}_{(c)}), {}^c C(n)_{n-1}/H_{n-1}({}^c C(n)_\bullet)).$$

Definition 2.15. Let

$${}^c \Theta_{\mathbf{D}}^{(n)} \in H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}_{(c)}), \mathbf{D}(\mathbb{X}, \mathbb{Z})^{(0)}/\delta_0).$$

be the class defined by pushing forward ${}^c \Theta^{S^{n-1}}$ via the map in Corollary 2.13.

3. EISENSTEIN CLASSES ON $(\mathbb{C}^\times)^n$

In this section, we construct $\mathrm{GL}_n(\mathbb{Z})$ -cohomology classes on algebraic tori, called “Eisenstein” or “polylogarithm” classes. These cohomological considerations will be used to connect our preceding explicit construction with later analytic calculations. We also hope that this algebraic material may also be motivational, helping to place our construction in the context of the literature.

3.1. Equivariant coherent cohomology. We introduce equivariant cohomology and review some of its formal properties, then discuss in greater detail the particular distributional Dolbeault model we will employ for our construction.

We will take a naive approach to (Borel) equivariant cohomology in this paper, as in Kings–Sprang [KS, Appendix A]; our treatment is identical to theirs,¹ so we just recall some facts. Let G be a discrete group, acting on the left on a complex manifold X with a closed G -submanifold $i : D \rightarrow X$, equipped with a G -sheaf $\mathcal{F}$.

Then the equivariant D -supported coherent cohomology $H_{G,D}^\bullet(X, \mathcal{F})$ is the left derived functor of the invariant global sections with support on D

$$\mathcal{F} \mapsto \Gamma_D(X, \mathcal{F})^G.$$

When $D = X$, the D is dropped from the notation and we simply call this the equivariant coherent cohomology with $\mathcal{F}$ -coefficients. The Grothendieck spectral sequence for a composite of functors gives the Hochschild–Serre spectral sequence

$$E_2^{p,q} = H^p(G, H_D^q(X, \mathcal{F})) \Rightarrow H_{G,D}^{p+q}(X, \mathcal{F})$$

and we also have a localization sequence (analogue of [KS, Definition A.3])

$$\dots \rightarrow H_{G,D}^i(X, \mathcal{F}) \rightarrow H_G^i(X, \mathcal{F}) \rightarrow H_G^i(X - D, \mathcal{F}) \rightarrow H_{G,D}^{i+1}(X, \mathcal{F}) \rightarrow \dots$$

In the case of varieties over $\mathbb{C}$, we will be interested in computing these groups analytically using the distributional Dolbeault resolutions of coherent sheaves (in particular, sheaves of

¹Unlike us, they work first in the *algebraic* category, not the analytic one, but formally things are very similar.

holomorphic differential n -forms). Recall that these sheaves $\mathcal{D}_X^{n,\bullet}$ are soft and thus their global sections compute coherent cohomology of Ω_X^n on X (i.e. derived functors of global sections); for the same general homological algebra reasons, the double complex given by forming group cochains $C^\bullet(G, \mathcal{D}_X^{n,\bullet})$ computes the equivariant coherent cohomology of Ω_X^n . Moreover, the localization sequence for a closed embedding of a smooth submanifold $D \hookrightarrow X$ is realized by the distinguished triangle

$$\mathcal{D}_{X,D}^{n,\bullet} \rightarrow \mathcal{D}_X^{n,\bullet} \rightarrow \mathcal{D}_{X-D}^{n,\bullet}$$

where the first term is the inclusion of currents supported on D ; there is a natural pushforward map $i_* : \mathcal{D}_D^{0,\bullet-n} \rightarrow \mathcal{D}_{X,D}^{n,\bullet}$ adjoint to restriction of compactly supported forms.

Remark 3.1. For the rest of this section, we will consider $\mathbf{T}$ (and its closed subschemes) as the complex manifold $(\mathbb{C}^\times)^n$ (and its closed complex submanifolds) by base change to $\mathbb{C}$ and analytification.

Now, observe that $\mathrm{GL}_n^+(\mathbb{Z})$ acts on the left on $\mathbf{T} - \{1\}$, and therefore on its space of currents by pushforward. We may compute $H_{\mathrm{GL}_n^+(\mathbb{Z})}^\bullet(\mathbf{T} - \{1\}, \Omega_{\mathbf{T}-\{1\}}^n)$ by the $\mathrm{GL}_n^+(\mathbb{Z})$ -equivariant cohomology of the complex

$$\mathcal{D}_{\mathbf{T}-\{1\}}^{n,0} \longrightarrow \mathcal{D}_{\mathbf{T}-\{1\}}^{n,1} \longrightarrow \mathcal{D}_{\mathbf{T}-\{1\}}^{n,2} \longrightarrow \dots,$$

which is the total cohomology of the double complex

$$C^\bullet(\mathrm{GL}_n^+(\mathbb{Z}), \mathcal{D}_{\mathbf{T}-\{1\}}^{n,\bullet}).$$

(In all of the above, and all that follows, we omit the global section functors from the soft sheaves of currents.)

We will also consider equivariant cohomology of the complements of some inductive systems consisting of unions of “toric hyperplanes”.

Definition 3.2. Let Z the set of group embeddings $\mathbb{G}_m^{n-1} \hookrightarrow \mathbf{T}$ translated by c -torsion sections which *do not* pass through $\{1\}$; this collection of subspaces is $\mathrm{GL}_n^+(\mathbb{Z})$ -invariant. Let $Z_\bullet = (Z_i)_{i \in I}$ be the ind-system of unions of elements in Z .

Observe that if $Z_{i_1}, Z_{i_2} \in Z_\bullet$ are such that $Z_{i_1} \subset Z_{i_2}$, then we have a map

$$A_{\mathbf{T}-Z_{i_2},c}^{n,\bullet} \longrightarrow A_{\mathbf{T}-Z_{i_1},c}^{n,\bullet}$$

defined by extending differential with compact support by zero from $\mathbf{T} - Z_{i_2}$ to $\mathbf{T} - Z_{i_1}$. Thus, we get a morphism

$$\mathcal{D}_{\mathbf{T}-Z_{i_1}}^{n,\bullet} \longrightarrow \mathcal{D}_{\mathbf{T}-Z_{i_2}}^{n,\bullet}.$$

In particular, we can consider the direct limit $\varinjlim_i \mathcal{D}_{\mathbf{T}-Z_i}^{n,\bullet}$.

Definition 3.3. We define $H_{\mathrm{GL}_n^+(\mathbb{Z})}^\bullet(\mathbf{T} - Z_\bullet, \Omega_{\mathbf{T}-Z_\bullet}^n)$ to be the cohomology of the double complex

$$C^\bullet(\mathrm{GL}_n^+(\mathbb{Z}), \varinjlim_i \mathcal{D}_{\mathbf{T}-Z_i}^{n,\bullet}).$$

Then the Hochschild–Serre spectral sequence in this pro-setting yields the following:

Theorem 3.4. *There exists a spectral sequence with second page*

$$E_2^{p,q} = H^p(\mathrm{GL}_n^+(\mathbb{Z}), \varinjlim_i H^q(\mathbf{T} - Z_i, \Omega_{\mathbf{T}-Z_i}^n)) \Rightarrow H_{\mathrm{GL}_n^+(\mathbb{Z})}^\bullet(T - Z_\bullet, \Omega_{\mathbf{T}-Z_\bullet}^n).$$

Moreover, since $\mathbf{T} - Z_i$ is Stein for every i , we have an isomorphism

$$H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), \varinjlim_i H^0(\mathbf{T} - Z_i, \Omega_{\mathbf{T}-Z_i}^n)) \xrightarrow{\sim} H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(T - Z_\bullet, \Omega_{\mathbf{T}-Z_\bullet}^n).$$

Proof. The spaces in the pro-system $\mathbf{T} - Z_\bullet$ are Stein, so the corresponding Hochschild–Serre spectral sequence has no terms in the second page with positive fiber degree. $\square$

3.2. Eisenstein classes on $\mathbf{T}$. We now construct equivariant classes for $\mathrm{GL}_n^+(\mathbb{Z})$ acting on the left on $\mathbf{T}$, analogous to the construction of [KS, §2] with their abelian scheme replaced by an algebraic torus (specialized to trivial coefficients instead of the big logarithm sheaf). These classes are usually called “Eisenstein” or “polylogarithm” classes in the literature, and our treatment of the construction below uses known methods. Our particular construction is most similar to the one in the “cas multiplicatif” of [BCG2, §1.3], though our stabilization (see further below) is different than theirs.

The long exact localization sequence of the closed $\mathrm{GL}_n^+(\mathbb{Z})$ -embedding of the identity $\{1\} \hookrightarrow \mathbf{T}$ yields

$$\dots \rightarrow H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(\mathbf{T}, \Omega_{\mathbf{T}}^n) \rightarrow H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(\mathbf{T} - \{1\}, \Omega_{\mathbf{T}-\{1\}}^n) \rightarrow H_{\mathrm{GL}_n^+(\mathbb{Z})}^0(\{1\}) \rightarrow H_{\mathrm{GL}_n^+(\mathbb{Z})}^n(\mathbf{T}, \Omega_{\mathbf{T}}^n) \rightarrow \dots$$

The scalar isogenies $[a]_* : \mathbf{T} \rightarrow \mathbf{T}$ act $\mathrm{GL}_n(\mathbb{Z})$ -equivariantly on all groups involved² so we may take trace-fixed parts of the sequence.

Proposition 3.5. *The sequence*

$$H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(\mathbf{T}, \Omega_{\mathbf{T}}^n)^{(0)} \rightarrow H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(\mathbf{T} - \{1\}, \Omega_{\mathbf{T}-\{1\}}^n)^{(0)} \rightarrow H_{\mathrm{GL}_n^+(\mathbb{Z}), \{1\}}^n(\mathbf{T})^{(0)} \rightarrow H_{\mathrm{GL}_n^+(\mathbb{Z})}^n(\mathbf{T}, \Omega_{\mathbf{T}}^n)^{(0)}$$

is exact, and moreover we have the following identifications:

- $H_{\mathrm{GL}_n^+(\mathbb{Z}), \{1\}}^n(\mathbf{T})^{(0)} = \mathbb{C}$, generated by the class of the identity current δ_1 .
- $H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(\mathbf{T}, \Omega_{\mathbf{T}}^n)^{(0)} \cong H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), H^0(\mathbf{T}, \Omega_{\mathbf{T}}^n))^{(0)} \cong H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), \mathbb{C})$. Moreover, $H^0(\mathbf{T}, \Omega_{\mathbf{T}}^n)^{(0)} \cong \mathbb{C}$ is generated by ω .
- $H_{\mathrm{GL}_n^+(\mathbb{Z})}^n(\mathbf{T}, \Omega_{\mathbf{T}}^n)^{(0)} \cong H^n(\mathrm{GL}_n^+(\mathbb{Z}), H^0(\mathbf{T}, \Omega_{\mathbf{T}}^n))^{(0)} \cong H^n(\mathrm{GL}_n^+(\mathbb{Z}), \mathbb{C})$.

Moreover, the image of the generator of $H_{\mathrm{GL}_n^+(\mathbb{Z})}^0(\{1\})^{(0)}$ is zero in $H_{\mathrm{GL}_n^+(\mathbb{Z})}^n(\mathbf{T}, \Omega_{\mathbf{T}}^n)^{(0)}$.

Proof. Algebraically, $\mathbf{T}$ is Stein, so its coherent cohomology with coefficients in $\Omega_{\mathbf{T}}^n$ is concentrated in degree zero, with $H^0(\mathbf{T}, \Omega_{\mathbf{T}}^n)$ freely generated as an $\mathcal{O}_{\mathbf{T}}$ -module by the form ω .

Let $M \subset \mathcal{O}_{\mathbf{T}}$ be the subspace of functions with vanishing constant Laurent coefficient. Then we claim that for a positive integer a , the $\mathrm{GL}_n(\mathbb{Z})$ -equivariant map

$$1 - [a]_* : \mathcal{O}_{\mathbf{T}}\omega \rightarrow \mathcal{O}_{\mathbf{T}}\omega,$$

²As customary, to define its action on cohomology groups of $\mathbf{T} - \{1\}$, we need to take the composite

$$H^\bullet(\mathbf{T} - \{1\}, \Omega^n) \rightarrow H^\bullet(\mathbf{T} - \mathbf{T}[a], \Omega^n) \rightarrow H^\bullet(\mathbf{T} - \{1\}, \Omega^n)$$

where the first arrow is pullback by the inclusion.

which kills $\mathbb{C}\omega$, restricts to an automorphism of M : indeed, its formal inverse

$$1 + [a]_* + [a]_*^2 + \dots$$

applied to $f\omega \in \mathcal{O}_T\omega$ for f with formal Laurent expansion

$$\sum_{\underline{m} \in \mathbb{Z}^n} c_{\underline{m}} z^{\underline{m}}.$$

Noting that $[a]_*$ turns the coefficient $c_{\underline{m}}$ into $c_{a\underline{m}}$, and that holomorphy implies $c_{\underline{m}}$ implies exponential decay in $\underline{m}$, we see that

$$(1 + [a]_* + [a]_*^2 + \dots)(f)$$

converges absolutely on compact balls to a holomorphic function.

One deduces that there exists an equivariant projection of $\Omega_{\mathbf{T}}^n$ onto M , and hence a splitting $\mathcal{O}_{\mathbf{T}}\omega \cong M \oplus \mathbb{C}\omega$ into the non-trace-fixed and trace-fixed parts. This equivariant splitting, applied to cohomology, then implies the second and third identifications, and also shows the exactness at the second place of the indicated trace-fixed part of the localization sequence.

The first listed identification requires another computation: the local cohomology classes is generated as infinite sums of the principal Laurent parts

$$\left[\frac{\omega}{z_1^{t_1} \dots z_n^{t_n}} \right]$$

for exponents $t_i \geq 1$. Calculating the pushforward on these germs, we obtain the diagonal action expression

$$[a]_* \left[\frac{\omega}{z_1^{t_1} \dots z_n^{t_n}} \right] = a^{t_1 + \dots + t_n} \left[\frac{\omega}{z_1^{t_1} \dots z_n^{t_n}} \right]$$

from which the first identification follows.

The exactness at the third place (requiring only any lifting of $[\delta_1]$), follow from the explicit lifts constructed in Section 3.3 below, and the last claim from the proof of [X1, Corollary 2.12], which produces an explicit lifting over the differential in the double complex for (a representative of) the image of $\{1\}$ in the equivariant coherent cohomology of $\mathbf{T}$. $\square$

The liftings of $\{1\}$ in the localization sequence therefore yield a subspace which we denote

$$\mathcal{E} \subset H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(\mathbf{T} - \{1\}, \Omega_{\mathbf{T}-\{1\}}^n)^{(0)},$$

which is a torsor under $H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), \mathbb{C})$ (realized inside the double complex via multiplying these cocycles by the trace-fixed determinant form ω).

Fix an auxiliary integer $c > 1$. Let $\mathbf{T}^\circ[c] \subset \mathbf{T}[c]$ be the non-identity c -torsion. We have the following variant of the construction by restricting to $\mathbf{T} - \mathbf{T}^\circ[c]$ instead of $\mathbf{T} - \mathbf{T}[1]$, proven exactly the same way:

Proposition 3.6. *The sequence*

$$H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(\mathbf{T}, \Omega_{\mathbf{T}}^n)^{(0_c)} \rightarrow H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(\mathbf{T} - \mathbf{T}[c], \Omega_{\mathbf{T}-\mathbf{T}[c]}^n)^{(0_c)} \rightarrow H_{\mathrm{GL}_n^+(\mathbb{Z})}^0(\mathbf{T}[c])^{(0_c)} \rightarrow H_{\mathrm{GL}_n^+(\mathbb{Z})}^n(\mathbf{T}, \Omega_{\mathbf{T}}^n)^{(0_c)}$$

is exact, and we have the identification

$$H_{\mathrm{GL}_n^+(\mathbb{Z})}^0(\mathbf{T}^\circ[c])^{(0_c)} = H^0(\mathbf{T}^\circ[c])^{\mathrm{GL}_n^+(\mathbb{Z})} \cong \mathbb{C}$$

mapping to zero in the sequence; the other two identifications of the preceding proposition hold identically with (0) replaced by (0_c) .

We denote by

$${}_c\mathcal{E} \subset H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(\mathbf{T} - \mathbf{T}[c], \Omega_{\mathbf{T}-\mathbf{T}^\circ[c]}^n)^{(0_c)},$$

the space of liftings of $[T[c] - \{1\}] \in H_{\mathrm{GL}_n^+(\mathbb{Z})}^0(\mathbf{T}^\circ[c])^{(0_c)}$. As before, this is a torsor under $H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), \mathbb{C})$.

Recall the ind-system $Z_\bullet$ introduced in Definition 3.2. From the definition of the cohomology groups via the corresponding double complexes, we observe that we have a restriction map

$$H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(\mathbf{T} - \mathbf{T}^\circ[c], \Omega_{\mathbf{T}-\mathbf{T}^\circ[c]}^n)^{(0_c)} \longrightarrow H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(\mathbf{T} - Z_\bullet, \Omega_{\mathbf{T}-Z_\bullet}^n)^{(0_c)}. \quad (3)$$

Moreover, Theorem 3.4 induces the isomorphism

$$H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(\mathbf{T} - Z_\bullet, \Omega_{\mathbf{T}-Z_\bullet}^n) \xrightarrow{\sim} H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), \varinjlim_i H^0(\mathbf{T} - Z_i, \Omega_{\mathbf{T}-Z_i}^n))^{(0_c)}.$$

Writing $\Omega_{\mathbf{T}-Z_\bullet}^n$ for the space of meromorphic n -forms on $\mathbf{T}$ with poles along hyperplanes in $Z_\bullet$, this affords us an identification

$$\varinjlim_i H^0(\mathbf{T} - Z_i, \Omega_{\mathbf{T}-Z_i}^n) = \Omega_{\mathbf{T}-Z_\bullet}^n.$$

Hence, we obtain

$$H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(\mathbf{T} - Z_\bullet, \Omega_{\mathbf{T}-Z_\bullet}^n)^{(0_c)} \cong H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), \Omega_{\mathbf{T}-Z_\bullet}^n)^{(0_c)}. \quad (4)$$

Thus, via the composition of (3) and (4), we obtain a map

$${}_c\mathcal{E} \longrightarrow H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), \Omega_{\mathbf{T}-Z_\bullet}^n)^{(0_c)}$$

realizing a $H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), \mathbb{C})$ -torsor inside this space.

Definition 3.7. We write

$${}_cZ^{(n)} \in H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), \Omega_{\mathbf{T}-Z_\bullet}^n / \mathbb{C}\langle\omega\rangle)^{(0_c)}$$

for the image of the torsor in the indicated group.

3.3. Double complex representatives. We now describe how equivariant cohomology classes are related (via spectral sequence edge maps) to group cohomology classes constructed from exact chain complexes via a lifting process, which will be the source of our cocycle constructions later. The idea is that the torsor ${}_c\mathcal{E}$ of lifts of $[\mathbf{T}[c] - \{1\}]$ is *characterized* by having this residue and being trace-fixed.

Definition 3.8. Let $Z \subset \mathbf{T}$ be a closed oriented submanifold of complex codimension s . We define the closed *current of integration* $\delta_Z \in \mathcal{D}_{\mathbf{T}}^{s,s}$ as follows. For $\eta \in \Omega_{\mathbf{T},c}^{n-s,n-s}$

$$\delta_Z(\eta) := \int_Z \eta.$$

Observe that reversing the orientation of Z turns δ_Z into $-\delta_{-Z}$.

The preceding discussion then has the following precise formulation:

Lemma 3.9. *Suppose that we have elements*

$$c_i \in C^{i-1}(\mathrm{GL}_n^+(\mathbb{Z}), (\mathcal{D}_{\mathbf{T}}^{n,n-i})^{(0_c)})$$

for $i = 1, 2, \dots, n$, such that the total differential of their sum $c = \sum c_i$ is $\sum_{x \in \mathbf{T}[c] \setminus \{1\}} \delta_x$. Then the pullback of c to $C^\bullet(\Gamma, \mathcal{D}_{\mathbf{T}-\mathbf{T}[c]}^{n,\bullet})$ represents a class in the torsor ${}^c\mathcal{E}$.

Such elements naturally arise from a lifting construction in group cohomology, recorded below in the form it is written in [X1, Lemma 2.5].

Proposition 3.10. *Suppose we have a $\mathrm{GL}_n^+(\mathbb{Z})$ -complex $C_\bullet$ supported in degrees $[0, n]$ exact except for having top homology $H_n(C_\bullet)$ free of rank 1. Suppose further that we have a map of complexes $r: C_\bullet \rightarrow (\mathcal{D}_{\mathbf{T}}^{n,n-\bullet})^{(0_c)}$, such that a generator of $H_n(C_\bullet)$ maps to a multiple of ω , and $e \in C_0$ maps to $\sum_{x \in \mathbf{T}[c] \setminus \{1\}} \delta_x$. Finally, suppose that the restriction of the image $r_n(C_n)$ lies in the image of*

$$(\Omega_{\mathbf{T}-Z_\bullet}^n)^{(0_c)} \hookrightarrow \mathcal{D}_{\mathbf{T}}^{n,0}.$$

Then the class in $H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), C_n/\mathbb{Z})$ coming from lifting $e \in C_0$ pushes forward under r_n to

$${}^cZ^{(n)} \in H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), \Omega_{\mathbf{T}-Z_\bullet}^n/\mathbb{C}\langle\omega\rangle)^{(0_c)}.$$

Proof. As in Lemma 2.14 and the discussion below it, the almost-exactness of $C_\bullet$ yields an element $c = c_1 + \dots + c_n \in C^\bullet(\mathrm{GL}_n^+(\mathbb{Z}), C_\bullet)$ whose total differential is $e \in C_0$ plus a contribution in $C^n(\mathrm{GL}_n^+(\mathbb{Z}), C_n)$ which is in the image of the $\mathbb{Z}$ -valued n -cocycles $Z^n(\mathrm{GL}_n^+(\mathbb{Z}), H_n(C_\bullet)) \cong Z^n(\mathrm{GL}_n^+(\mathbb{Z}), H_n(C_\bullet))$. (This comes from applying Lemma 2.14 to the one-term-longer complex where we add in the first kernel, $\mathbb{Z} \rightarrow C_\bullet$.)

The latter component is then a representative of the transgression cohomology class coming from the complex considered in [X1, (2.13)], which is shown there to be the $\mathrm{GL}_n^+(\mathbb{Z})$ -equivariant Euler class of S^{n-1} – and in particular, vanishes rationally. In particular, as in Corollary 2.12 of loc. cit., there exists a (non-unique) element

$$\phi \in C^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), \mathbb{C}\langle\omega\rangle)$$

for which

$$d_{\mathrm{tot}}\phi + r_*c$$

satisfies the conditions of Lemma 3.9, and hence represents an element in the torsor ${}^c\mathcal{E}$. We now focus on $[c_n]$, whose projection to be valued in $C_n/\mathbb{Z}$ represents the lifted class in $H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), C_n/\mathbb{Z})$. Since the restriction of $(r_n)_*(d_{\mathrm{tot}}\phi + c_n)$ to $T - Z_\bullet$ is closed along $T - Z_\bullet$, by [X, Lemma 2.2], this component yields the image under the Hochschild–Serre edge map

$$H_{\mathrm{GL}_n^+(\mathbb{Z})}^{n-1}(\mathbf{T} - Z_\bullet, \Omega_{\mathbf{T}-Z_\bullet}^n) \rightarrow H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}), \Omega_{\mathbf{T}-Z_\bullet}^n)$$

for the class $[d_{\mathrm{tot}} + r_*c]$. Thus, its projection to $(\Omega_{\mathbf{T}-Z_\bullet}^n)/\mathbb{C}\langle\omega\rangle$, which is now independent of ϕ , then must yield the class ${}^cZ^{(n)}$, as desired. $\square$

In particular, the complex ${}^cC(n)_\bullet$, together with the map

$${}^cC(n)_{\bullet-1} \xrightarrow{{}^c\Psi} \mathcal{D}_{\mathbf{T}}^{n,\bullet} \tag{5}$$

satisfies all the conditions of Proposition 3.10, and hence the liftings within this complex realize $\mathrm{GL}_n^+(\mathbb{Z})$ -cocycles representing ${}^c Z^{(n)}$:

If we write $\mathcal{O}_{(c)}$ for the $\mathbb{Z}$ -span of the image of Ψ in top degree, we then have:

Corollary 3.11. *The cocycle ${}^c \Theta^{(n)}$ agrees with ${}^c Z^{(n)}$ under the inclusion*

$$\mathcal{O}_{(c)}/\mathbb{Z} \hookrightarrow (\Omega_{\mathbf{T}-\mathbf{Z}_\bullet}^n)^{(0_c)}/\mathbb{C}\langle\omega\rangle.$$

As mentioned previously, we will make use of this for some analytic computations in the next section.

4. ANALYTIC REPRESENTATIVES OF THE EISENSTEIN CLASS

In this section, we construct a canonical analytic representative of our equivariant coherent class, similar to [KS, §3]. The Fourier-dual approach used there is less amenable in our setting since algebraic tori are not compact, so we follow instead the construction of Bergeron–Charollois–Garcia [BCG20], [BCG2], but replacing the Mathai–Quillen transgression form with a “Bochner–Martinelli form” adapted to the holomorphic structure. In particular, our construction is essentially a “Dolbeault version” of the theta series underlying the “cocycle multiplicatif” in [BCG2, Chapitre 8], and pushes forward to its class along the Hodge–de Rham edge map.

It will result that integrals of this series can be used to compute specializations of our Eisenstein class, and compute that the corresponding periods along nonsplit tori give L -values attached to abelian extensions of totally real fields (analogous to the calculation in [BCG20, §12]).

4.1. Geometric model for equivariant cohomology. Let $V := \mathbb{C}^n$ and $S := \mathrm{GL}_n^+(\mathbb{R})/\mathrm{SO}(n)$, both equipped with the $\mathrm{GL}_n(\mathbb{R})$ -action given by left multiplication. Consider also $\mathbf{T} = V/\mathbb{Z}^n$ which admits an action of $\mathrm{GL}_n^+(\mathbb{Z})$,

Consider the projection maps

$$\mathrm{pr}_S: S \times \mathbf{T} \longrightarrow S, \quad \mathrm{pr}_{\mathbf{T}}: S \times \mathbf{T} \longrightarrow \mathbf{T}$$

and define

$$\begin{aligned} \mathcal{E}^{1,0} &:= \mathrm{pr}_{\mathbf{T}}^* A_{\mathbf{T}}^{1,0}, \quad \mathcal{E}^{0,1} := \mathrm{pr}_S^* (A_X^1 \otimes \mathbb{C}) \oplus \mathrm{pr}_{\mathbf{T}}^* A_{\mathbf{T}}^{0,1}, \\ \mathcal{E}^{p,q} &:= \wedge^p \mathcal{E}^{1,0} \otimes \wedge^q \mathcal{E}^{0,1}, \end{aligned}$$

where the wedge product is over $C^\infty(S \times \mathbf{T})$. The differential on $\mathcal{E}^{0,\bullet}$ is given by $d_{tot} := d_S + \bar{\partial}_V$. For an open set U , we define $\mathcal{E}_c^{0,q}(U)$ to be the form in $\mathcal{E}^{0,q}(U)$ of compact support, which also forms a complex via d_{tot} .

Note that S has dimension $m = n(n+1)/2$, and therefore $S \times \mathbf{T}$ has dimension $n(n+1)/2 + 2n$. Let $s = m + n$. For $U = \mathbf{T} - D$ an open set, we consider smooth currents

$$\mathcal{D}_{S \times U}^r := \mathrm{Hom}_{cts}(\mathcal{E}_c^{0,s-r}(S \times U), \mathbb{C})$$

with the appropriate topology and differential d_{tot} induced by the differential on $\mathcal{E}_c^{0,s-\bullet}(U)$. We then get a complex

$$\mathcal{D}_{S \times U}^\bullet = (\mathcal{D}_{S \times U}^0 \longrightarrow \mathcal{D}_{S \times U}^1 \longrightarrow \dots \longrightarrow \mathcal{D}_{S \times U}^s).$$

Proposition 4.1. *Let Γ be a finite index subgroup of $\mathrm{GL}_n^+(\mathbb{Z})$, and denote by $U = \mathbf{T} - D$ an open set that is Γ -invariant. The complex of Γ -invariants $(\mathcal{D}_{S \times U}^\bullet)^\Gamma$ computes the equivariant coherent cohomology groups $H_\Gamma^\bullet(\mathbf{T} - D, \Omega_{\mathbf{T}-D}^n)$.*

In this section, we will construct distinguished elements in $(\mathcal{D}_{S \times \mathbf{T}}^{n-1})^{(0)}$ whose total differential is supported on some torsion sections, which will represent (a slight transformation of) the torsor ${}^c\mathcal{E} \subset H_\Gamma^{n-1}(\mathbf{T} - \mathbf{T}^\circ[c], \Omega_{\mathbf{T}-\mathbf{T}^\circ[c]}^n)^{(0)}$. We will use the regularized theta-summation method employed in [BCG20] and [BCG2]. More precisely, we will first construct differential forms on $X \times (V - \{0\})$ with rapid decay along the fibers and differential equal to a current of integration along the zero section. Then, via a regularized theta summation, we will construct Γ -invariant differential forms on $S \times (\mathbf{T} - D)$ which will map to the complex of distributions we described.

4.2. A Thom form and the transgression form. In this section, we will work on the space $S \times V$. Define $\mathcal{E}_{S \times V}^{p,q}$ similarly as before, but replacing $\mathbf{T}$ by V . If it is clear from the context, we will sometimes remove the subindex $S \times V$. In this section, we construct differential form

$$\eta \in \mathcal{E}_{S \times V}^{n,n-1}(X \times V - \{0\})$$

that is locally integrable along the zero section, so that it determines a distribution on $S \times V$ whose differential is equal to $\delta_{S \times \{0\}}$.

Represent a point in S by a positive-definite matrix representative via the bijection

$$S \xrightarrow{\sim} \{H \mid H = H^T \text{ and } H \text{ positive-definite}\}, [h] \mapsto H = h^{-T} h^{-1},$$

and set

$$\mathbf{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} := H\bar{\mathbf{z}},$$

where $\mathbf{z}$ is the coordinate on V . Write $\omega := dz_1 \wedge \dots \wedge dz_n$ for the fiberwise determinant form. We define

$$\varphi := \kappa_n e^{-\pi Q_H(\mathbf{z})} d_{\text{tot}} u_1 \wedge \dots \wedge d_{\text{tot}} u_n \wedge \omega \in \mathcal{E}_{rd}^{n,n}(X \times V), \quad \text{where } \kappa_n = \frac{(-1)^{n(n-1)/2}}{(2i)^n},$$

where the subindex rd means rapid decay along the fibers and $Q_H(z) = z^T H \bar{z}$. We now show some properties about φ .

Proposition 4.2. *The element φ is closed, equivariant under the action of $\mathrm{GL}_n(\mathbb{R})$ and its integral along every fiber is equal to 1.*

Let $X^{0,1} := \sum \bar{z}_i \partial \bar{z}_i$ be the antiholomorphic radial vector field and consider

$$\psi := \iota_{X^{0,1}} \varphi = \kappa_n e^{-\pi Q_H(\mathbf{z})} \Omega(H, z) \wedge \omega \in \mathcal{E}_{rd}^{n,n-1}(X \times V),$$

where

$$\Omega(H, z) = \sum_{j=1}^n (-1)^{j-1} u_j d_{\text{tot}} u_1 \wedge \dots \widehat{d_{\text{tot}} u_j} \dots \wedge d_{\text{tot}} u_n. \quad (6)$$

and the equality follows from computing $d_{tot}\mathbf{u} = H d\bar{\mathbf{z}} + (d_\Gamma H)\bar{\mathbf{z}}$ where the latter term is killed by the antiholomorphic contraction.

For $t \in \mathbb{R}_{\geq 0}$, consider the map $[t]: X \times V \rightarrow X \times V$ given by multiplication by t on the fiber. We finally give the transgression form η .

Proposition 4.3. *Consider the element*

$$\eta := 2 \int_0^{+\infty} [t]^* \psi \frac{dt}{t}.$$

It defines a form $\eta \in \mathcal{E}^{n,n-1}(X \times V - \{0\})$ that is locally integrable along the zero section. The corresponding current on $X \times V$ satisfies $d_{tot}\eta = \delta_{X \times 0} \in \mathcal{D}^n(V)$.

For future calculations, it will be convenient to calculate an explicit formula for η and for the following modification of it.

Proposition 4.4. *For $s \in \mathbb{C}$ with $\operatorname{Re}(s) \geq 0$, consider*

$$\eta(s) := 2 \int_0^{+\infty} [t]^* \psi t^s \frac{dt}{t}.$$

Then,

$$\eta(s) = \frac{\kappa_n \Gamma(n + s/2)}{\pi^{n+s/2}} \frac{\Omega(H, z) \wedge \omega}{Q_H(z)^{n+s/2}}.$$

4.3. Eisenstein transgression. We now consider a regularized average of the form η over a lattice. For every $\lambda \in V$ define

$$\operatorname{tr}_\lambda: X \times V \longrightarrow X \times V, (g, v) \longmapsto (g, v + \lambda).$$

For a Schwartz function $\phi \in \mathcal{S}(V_{\mathbb{A}_f})$, we consider

$$\theta(\psi, \phi) := \sum_{\lambda \in V_{\mathbb{Q}}} \phi(\lambda) \operatorname{tr}_{-\lambda}^* \psi.$$

In fact, we will mostly be interested in the following generalization of the previous expression. For $t \in \mathbb{R}_{>0}$

$$\theta([t]^* \psi, \phi) := \sum_{\lambda \in V_{\mathbb{Q}}} \phi(\lambda) \operatorname{tr}_{-\lambda}^* [t]^* \psi.$$

Consider $D := \sum_x a_x(x)$ a torsion cycle in $\mathbf{T} \simeq (\mathbb{C}/\mathbb{Z})^n$ of degree 0 which is Γ -invariant, for Γ a finite index subgroup of $\operatorname{SL}_n(\mathbb{Z})$. For every x in the support of the divisor, choose a representative $\tilde{x} \in \mathbb{Q}^n$. We can then consider the Schwartz function

$$\phi_D := \sum_x a_x 1_{\tilde{x} + \hat{\mathbb{Z}}^n}, \text{ where } 1_{\tilde{x} + \hat{\mathbb{Z}}^n} = \bigotimes_q 1_{\tilde{x} + \mathbb{Z}_q^n}.$$

Denote by $L_D \subset V$ the support of ϕ_D when restricted to $\mathbb{Q}^n$.

Theorem 4.5. *Suppose that D is of degree 0 and view $\theta([t]^* \psi, \phi_D)$ as a differential form on $X \times (V - L_D)$. For $s \in \mathbb{C}$ with $\operatorname{Re}(s) \gg 0$, the integral*

$$E_\psi(\phi_D, s) := 2 \int_0^{+\infty} \theta([t]^* \psi, \phi_D) t^s \frac{dt}{t}$$

converges. Furthermore, it admits a meromorphic continuation at all $s \in \mathbb{C}$, regular at $s = 0$ and its value at every regular s defines a Γ -invariant smooth form on $\mathcal{E}^{n,n-1}(X \times (V - L_D))$. At $s = 0$, the form on $X \times (V - L_D)$ is closed.

We will denote $E_\psi(\phi_D) := E_\psi(\phi_D, 0)$. It follows from the expression of $E_\psi(\phi_D)$ as a regularized theta series that $E_\psi(\phi_D)$ descends to an element in $\mathcal{E}^{n,n-1}(X \times (\mathbf{T} - D))^\Gamma$.

Theorem 4.6. *Suppose that D is of degree 0. The form $E_\psi(\phi_D) \in \mathcal{E}^{n,n-1}(X \times (\mathbf{T} - D))$ is locally integrable along D . The corresponding distribution in $(\mathcal{D}_{X \times \mathbf{T}}^{n-1})$ has differential given by*

$$d_{\text{tot}}(E_\psi(\phi_D)) = \delta_D.$$

4.4. Comparison of cohomology classes. We now specialize the discussion to the kind of torsion cycles that will be relevant for our construction. Recall that c is a prime that is sufficiently large. In addition, let d be prime to c . Consider a finite order nontrivial character

$$\chi: \mathbf{T}[d] \longrightarrow \mathbb{C}^\times.$$

It determines a compact open set $V_\chi \subset \mathbb{X}_0$, which has stabilizer $\Gamma_\chi \subset \text{GL}_n^+(\mathbb{Z})$. Consider now the cycle

$$D_\chi = \frac{1}{d^n} \sum_{x \in \mathbf{T}[d]} \chi(x)^{-1}(x) \in \mathbb{C}[\mathbf{T}[d]].$$

Let $K = \mathbf{T}[cd] - \mathbf{T}[d]$. Observe that translation by $x \in \mathbf{T}[d]$ induces a map

$$t_x: \mathbf{T} - K \longrightarrow \mathbf{T} - K, \quad a \longmapsto a + x.$$

While the individual maps t_x are not Γ_χ -equivariant, the fact that the divisor D is Γ_χ -invariant implies that the following weighted sum of pullbacks induces a map in equivariant coherent cohomology

$$t_{D_\chi}^* := \frac{1}{d^n} \sum_x \chi(x)^{-1} t_x^*: H_{\Gamma_\chi}^{n-1}(\mathbf{T} - K, \Omega_{\mathbf{T}-K}^n) \longrightarrow H_{\Gamma_\chi}^{n-1}(\mathbf{T} - K, \Omega_{\mathbf{T}-K}^n).$$

Lemma 4.7. *Consider the same notation as above. We have:*

- (1) *The map $t_{D_\chi}^*$ is an idempotent.*
- (2) *The image of $w \in H_\Gamma^{n-1}(\mathbf{T}, \Omega_{\mathbf{T}}^n)^{(0,c)}$ in $H_{\Gamma_\chi}^{n-1}(\mathbf{T} - K, \Omega_{\mathbf{T}-K}^n)$ is in the kernel of $t_{D_\chi}^*$. In particular, $t_{D_\chi}^*$ can be viewed as a map*

$$t_{D_\chi}^*: H_{\Gamma_\chi}^{n-1}(\mathbf{T} - K, \Omega_{\mathbf{T}-K}^n) / H^{n-1}(\Gamma_\chi, \mathbb{C}w) \longrightarrow H_{\Gamma_\chi}^{n-1}(\mathbf{T} - K, \Omega_{\mathbf{T}-K}^n).$$

Definition 4.8. Denote by $t_{D_\chi}^*(c\mathcal{E}) \in H_{\Gamma_\chi}^{n-1}(\mathbf{T} - K, \Omega_{\mathbf{T}-K}^n)$ the image of the restriction of the torsor $c\mathcal{E} \subset H_{\Gamma_\chi}^{n-1}(\mathbf{T} - \mathbf{T}^\circ[c], \Omega_{\mathbf{T}-\mathbf{T}^\circ[c]}^n)$ to $\mathbf{T} - K$.

The next theorem uses the Eisenstein series from the previous section to give an analytic representative of the class $t_{D_\chi}^*(c\mathcal{E})$. For that, we need to set some notation. Let $\phi_c = 1_{(1/c)\mathbb{Z}^n} - 1_{\mathbb{Z}^n} \in S(V_{\mathbb{Q}})$ and define $\phi_{D_\chi, c} \in \mathcal{S}(V_{\mathbb{Q}})$ as

$$\phi_{D_\chi, c}(v) := \frac{1}{d^n} \sum_x \chi(x)^{-1} \phi_c(v + \tilde{x}),$$

where $\tilde{x} \in \mathbb{Q}^n$ is a lift of $x \in \mathbf{T}[d]$.

Theorem 4.9. *The element $E_\psi(\phi_{D_{\chi,c}})$ defines a closed element in $(\mathcal{D}_{\mathbf{T}-K}^{n-1})^{\Gamma_\chi}$ whose cohomology class is equal to $t_{D_\chi}^*(c\mathcal{E}) \in H_{\Gamma_\chi}^{n-1}(\mathbf{T} - K, \Omega_{\mathbf{T}-K}^n)$.*

Corollary 4.10. *Denote by $e: X \rightarrow X \times \mathbf{T}$ the zero section and by ι_w contraction along the vector field $\partial/\partial x_1 \otimes \cdots \otimes \partial/\partial x_n$. Let $z \in X$ be an arbitrary point. Then, the map*

$$\Gamma_\chi^n \longrightarrow \mathbb{C}, (\gamma_0, \dots, \gamma_{n-1}) \longmapsto \int_{\Delta(\gamma_0 z, \dots, \gamma_{n-1} z)} e^* \iota_w(E_\psi(\phi_{D_{\chi,c}}))$$

defines a cocycle that represents the class $H^{n-1}(\Gamma_\chi, \mathbb{C})$ given by evaluation at the open set V_χ the class ${}_c\Theta^{\mathbf{D}}$ (after extending scalars to $\mathbb{C}$).

More generally, we can consider a compact open $V = \cup_{\chi \in I} V_{\chi_i}$ for several non-trivial characters $\{\chi_i\}_{i \in I} \subset \text{Hom}(\mathbf{T}[d], \mathbb{C}^\times)$. Denote by $\Gamma_V \subset \text{GL}_n^+(\mathbb{Z})$ the stabilizer of V . We can then define $D_V = \sum_{i \in I} D_{\chi_i}$ and

$$t_{D_V}^* := \sum_{\chi} t_{D_{\chi_i}}^* : H_{\Gamma_V}^{n-1}(\mathbf{T} - K, \Omega_{\mathbf{T}-K}^n) \longrightarrow H_{\Gamma}^{n-1}(\mathbf{T} - K, \Omega_{\mathbf{T}-K}^n).$$

The orthogonality of characters implies the same assertions of the lemma above also hold for t_D^* . On the other hand, let $\phi_{V,c} = \sum_{i \in I} \phi_{D_{\chi_i},c}$.

Then, we have the following generalization of the corollary above.

Corollary 4.11. *Denote by $e: X \rightarrow X \times \mathbf{T}$ the zero section and by ι_w contraction along the vector field $\partial/\partial x_1 \otimes \cdots \otimes \partial/\partial x_n$. Let $z \in X$ be an arbitrary point. Then, the map*

$$\Gamma_V^n \longrightarrow \mathbb{C}, (\gamma_0, \dots, \gamma_{n-1}) \longmapsto \int_{\Delta(\gamma_0 z, \dots, \gamma_{n-1} z)} e^* \iota_w(E_\psi(\phi_{V,c}))$$

defines a cocycle that represents the class $H^{n-1}(\Gamma_V, \mathbb{C})$ given by evaluation at the open set V the class ${}_c\Theta^{\mathbf{D}}$ (after extending scalars to $\mathbb{C}$).

4.5. Restriction to tori attached to a totally real field and L -values. Let F be a totally real field of degree n over $\mathbb{Q}$. Let $\mathfrak{a}$ be an integral ideal of F . Let $\tau \in F^n$ be a $\mathbb{Z}$ -basis of $\mathfrak{a}^{-1}$. Then, it induces an isomorphism

$$\beta: \mathbb{Z}^n \xrightarrow{\sim} \mathfrak{a}^{-1}, x \longmapsto \tau^t \cdot x.$$

It also induces an injection

$$\iota_\tau: F \hookrightarrow M_n(\mathbb{Q}), \alpha \longmapsto A_\alpha,$$

where A_α is determined by $\alpha(\tau^t \cdot x) = \tau^t \cdot A_\alpha x$. From this last equality, we deduce

$$A_\alpha^t \tau = \alpha \tau.$$

From there, and using $A_\alpha \in M_n(\mathbb{Q})$, we have that A_α^t diagonalizes with eigenvectors $\sigma_i(\tau)$, where $\{\sigma_i\}$ denotes the set of embeddings $F \hookrightarrow \mathbb{R}$. Let $g \in \text{GL}_n(\mathbb{R})$ be the change of basis matrix

$$A_\alpha = g^{-1} D_\alpha g,$$

so that D_α is diagonal. Let $\delta = \det g > 0$. If $\tau = (\tau_i)_i \in F^n$, we denote by $\tau^\vee = (\tau_j^\vee)$ its dual basis, so that

$$\text{Tr}_{F/\mathbb{Q}}(\tau_i \tau_j^\vee) = \delta_{ij}.$$

Let

$$\beta^\vee: \mathbb{Q}^n \xrightarrow{\sim} F, \quad v \mapsto \sum \tau_j^\vee v_j.$$

Consider now the inclusion

$$j_\tau: (F \otimes \mathbb{R})^{1,+} \longrightarrow \mathcal{X} = \mathrm{SL}_n(\mathbb{R})/\mathrm{SO}_n, \quad \lambda \mapsto [\delta^{-1/n} g^t D_\lambda^{-1}]$$

Denote by $X(F)$ the image of $(F \otimes \mathbb{R})^1$ in $\mathcal{X}$. Observe that using that the matrices in this image diagonalize, we can fix coordinates of the form

$$\{(\lambda_1, \dots, \lambda_n) \mid \lambda_i \in \mathbb{R}_{>0} \text{ such that } \prod_i \lambda_i = 1\} \mapsto [\delta^{-1/n} g^t D_\lambda^{-1}],$$

where $D_\lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$. Then, the corresponding positive definite matrix would be

$$\delta^{2/n} g^{-1} D_\lambda^2 g^{-t}.$$

On $\mathbb{C}^n$, we will put the coordinates

$$\zeta = \delta^{1/n} g^{-t} \mathbf{z}.$$

Lemma 4.12. *Let $\varphi_i := (1/2i)e^{-\pi\lambda_i^2|\zeta_i|^2} d_{\mathrm{tot}}(\lambda_i^2 \bar{\zeta}_i) \wedge d\zeta_i$ and $\psi_i := (1/2i)e^{-\pi\lambda_i^2|\zeta_i|^2} \lambda_i^2 \bar{\zeta}_i d\zeta_i$. Then, form φ and ψ restricted to $X_F \times V$ can be written as*

$$\iota^* \varphi = \varphi_1 \wedge \dots \wedge \varphi_n,$$

$$\iota^* \psi = \sum_{i=1}^n \varphi_1 \wedge \dots \wedge \varphi_{i-1} \wedge \psi_i \wedge \dots \wedge \varphi_n.$$

Now, let $\alpha \in F$ and denote by $v_\alpha = (\beta^\vee)^{-1}(\alpha) \in \mathbb{Q}^n$. Consider the section

$$s_\alpha: X(F) \longrightarrow \mathcal{X} \times V, \quad \lambda \mapsto (j_\tau(\lambda), v_\alpha).$$

Consider the restriction of the differential form $\eta(s) \in \mathcal{E}_{X \times V}^{n, n-1}$ to $\mathcal{X} \times V$, that we will denote by the same symbol.

Lemma 4.13. *We have*

$$I_F(\alpha, s) := \int_{X(F)} s_\alpha^* \iota_\omega \eta(s) = \frac{\kappa_n \cdot \delta^{-1-s/n}}{\pi^{n+s/2}} \Gamma\left(1 + \frac{s}{2n}\right)^n \frac{1}{N_{F/\mathbb{Q}}(\alpha) |N_{F/\mathbb{Q}}(\alpha)|^{s/n}}.$$

Let $f \in \mathbb{Z}$ and denote by $U_F(f)$ the totally positive units in $\mathcal{O}_F$ that are congruent to 1 modulo f .

Proposition 4.14. *Let $\phi \in \mathcal{S}(V_{\mathbb{A}_f})$ be a Schwartz function satisfying:*

- (1) ϕ is $\rho(U_F(f))$ -equivariant, where $\rho: U_F(f) \rightarrow \mathrm{SL}_n(\mathbb{Z})$ is defined as $\rho(u) = A_u^{-T}$.
- (2) $\phi(v + \ell) = \phi(v)$ for every $\ell \in \hat{\mathbb{Z}}^n$,
- (3) $\phi(0) = 0$,
- (4) $\hat{\phi}(0) = \int_{\mathbb{A}_f^n} \phi(v) dv = 0$, where dv is a Haar measure for which $\hat{\mathbb{Z}}^n$ has total measure 1.

Then,

$$\int_{U_F(f) \backslash (F \otimes \mathbb{R})^1} j_\tau^* e^* \iota_W E_\psi(\phi) = \frac{\kappa_n}{\pi^n \delta} Z_f(1) = \frac{\kappa_n (-i)^n}{\delta} Z_{\hat{f}}(0),$$

where $f(y) = \phi((\beta^\vee)^{-1}(y))$, $\hat{f}$ is the Fourier transform of f and

$$Z_f(s) = \sum_{y \in U_F(f) \backslash F^\times} f(y) \frac{\text{sgn} N(y)}{|N(y)|^s}$$

for $\text{Re}(s) \gg 0$ and is extended to all s by analytic continuation.

Proof. Observe that, as a consequence of defining $E_\psi(\phi)$ as a regularized theta sum,

$$j_\tau^* e^* \iota_W E_\psi(\phi, s) = \sum_{y \in F^\times} f(y) s_y^* \iota_W \eta(s).$$

Now, using that f is invariant under $\rho(U_F(f))$, we can rewrite the integral as

$$\int_{U_F(f) \backslash (F \otimes \mathbb{R})^1} \sum_{y \in F^\times} f(y) s_y^* \iota_W \eta(s) = \sum_{y \in U_F(f) \backslash F^\times} f(y) \int_{(F \otimes \mathbb{R})^1} s_y^* \iota_W \eta(s) = \frac{\kappa_n}{\pi^n \delta} Z_f(1),$$

where we used Lemma 4.13 and the definition of $Z_f(s)$ in the last equality. The second equality follows from the functional equation

$$\gamma(s)^n Z_f(s) = (-i)^n \gamma(1-s)^n Z_{\hat{f}}(1-s),$$

where $\gamma(s) = \pi^{-(s+1)/2} \Gamma((s+1)/2)$. $\square$

4.6. Interpretation in terms of the measure-valued class. Let S be a set of primes of F containing p , all primes dividing f and prime to c . Let $S_{\mathbb{Q}}$ be the set of primes of $\mathbb{Q}$ below the places of S , and let S' be the set of places of F above $S_{\mathbb{Q}}$, so in particular $S \subset S'$. Consider

$$\mathcal{U} := \prod_{\substack{v \in S \\ v \nmid p}} U_{v,f}, \text{ where } U_{v,f} := \{x \in \mathcal{O}_v^\times \mid x \equiv 1 \pmod{f \mathcal{O}_v}\}.$$

Let $\mathbb{O} \subset \mathcal{O}_p$ be either equal to $\mathcal{O}_p$ or to $\mathcal{O}_p^\times$. The inverse of the map β induces an inclusion

$$\mathbb{O} \times \mathcal{U} \times \prod_{v \in S' - S} \mathcal{O}_v \xrightarrow{\beta^{-1}} \left(\prod_{q \in S_{\mathbb{Q}}} \mathbb{Z}_q^n \right)',$$

where the prime at the right-hand side denotes primitive vectors. Since S is prime to c , we can consider the following maps of $\mathbb{Z}$ -valued measures

$$\mathbf{D}(\mathbb{X}_0)/\delta_0 \longrightarrow \mathbf{D} \left(\prod_{q \in S_{\mathbb{Q}}} \mathbb{Z}_q^n \right) / \delta_0 \longrightarrow \mathbf{D} \left(\left(\prod_{q \in S_{\mathbb{Q}}} \mathbb{Z}_q^n \right)' \right) \longrightarrow \mathbf{D}(\mathbb{O} \times \mathcal{U} \times \prod_{v \in S' - S} \mathcal{O}_v) \longrightarrow \mathbf{D}(\mathbb{O} \times \mathcal{U}),$$

where the first map is given by pushforward, the second by restriction, the third by restriction and the identification induced by β , and the last one by pushforward again.

Let $\mathfrak{m}$ be an ideal of F supported on the primes dividing S , let $U_{\mathfrak{m}} := \{x \in \mathbb{O} \times \mathcal{U} \mid x \equiv 1 \pmod{\mathfrak{m}}\}$. Then, for every $\alpha \in \mathbb{O} \times \mathcal{U} / (U_{\mathfrak{m}} U_F(f))$, we determine the open set $U_\alpha \subset \mathbb{O} \times \mathcal{U}$ as the pre-image of α via the quotient map

$$\mathbb{O} \times \mathcal{U} \longrightarrow \mathbb{O} \times \mathcal{U} / (U_{\mathfrak{m}} U_F(f)).$$

Denote by $G_{\mathfrak{f}}$ the narrow ray class group of F of modulus $\mathfrak{f} = f\mathcal{O}$, and define similarly $G_{\mathfrak{m}}$. Since $\mathfrak{f} \mid \mathfrak{m}$, consider the projection map

$$G_{\mathfrak{m}} \twoheadrightarrow G_{\mathfrak{f}}$$

and let $H_{[\mathfrak{a}]_{\mathfrak{f}}, \mathfrak{m}}$ be the pre-image of $[\mathfrak{a}]_{\mathfrak{f}}$ in $G_{\mathfrak{m}}$. Observe that we have a bijection

$$\mathbb{O} \times \mathcal{U}/(U_{\mathfrak{m}}U_F(f)) \xrightarrow{\sim} H_{[\mathfrak{a}]_{\mathfrak{f}}, \mathfrak{m}}, \quad [\alpha] \mapsto [(\tilde{\alpha})\mathfrak{a}]_{\mathfrak{m}},$$

where on the right-hand side, $\tilde{\alpha} \in \mathcal{O}_F$ denotes a totally positive element whose reduction modulo $\mathfrak{m}$ equals to α . Finally, for an ideal class $[\mathfrak{b}]_{\mathfrak{m}} \in G_{\mathfrak{m}}$, let

$$L_S(H_{\mathfrak{m}}/F, [\mathfrak{b}]_{\mathfrak{m}}, s) = \sum_{\substack{(I, S)=1 \\ [I]_{\mathfrak{m}}=[\mathfrak{b}]_{\mathfrak{m}}}} NI^{-s},$$

and consider the c -smoothed L -value

$$L_{S,c}(H_{\mathfrak{m}}/F, [\mathfrak{b}]_{\mathfrak{m}}, 0) := L_S(H_{\mathfrak{m}}/F, [\mathfrak{b}]_{\mathfrak{m}}, 0) - c^n L_S(H_{\mathfrak{m}}/F, [(c)^{-1}\mathfrak{b}]_{\mathfrak{m}}, 0).$$

Theorem 4.15. *Consider the same notation as above. If $\mathbb{O} = \mathcal{O}_p^{\times}$, we have*

$${}_c\Theta^{\mathbf{D}}(c_{U_F(f)})(U_{\alpha}) = -(-1)^{n(n+1)/2} L_{S,c}(H_{\mathfrak{m}}/F, [(\tilde{\alpha})\mathfrak{a}]_{\mathfrak{m}}, 0).$$

If on the other hand $\mathbb{O} = \mathcal{O}_p$ and $f > 1$,

$${}_c\Theta^{\mathbf{D}}(c_{U_F(f)})(U_{\alpha}) = -(-1)^{n(n+1)/2} L_{R,c}(H_{\mathfrak{m}}/F, [(\tilde{\alpha})\mathfrak{a}]_{\mathfrak{m}}, 0),$$

where $R = S - \{p\mathcal{O}\}$.

Proof. We prove the first equality, as the second one is analogous. Via the identifications explained above, the open set U_{α} corresponds to the following open sets. First, let

$$W_{\alpha} := \beta^{-1} \left(U_{\alpha} \times \prod_{v \in S' - S} \mathcal{O}_v \right) \subset \prod_{q \in S_{\mathbb{Q}}} \mathbb{Z}_q^n.$$

Then, if we consider the natural projection

$$\pi_{S_{\mathbb{Q}}} : \mathbb{X}_0 \twoheadrightarrow \prod_{q \in S_{\mathbb{Q}}} \mathbb{Z}_q^n,$$

let

$$V_{\alpha} := \pi_S^{-1}(W_{\alpha}) \subset \mathbb{X}_0 - \{0\}.$$

Choose N supported in the primes in $S_{\mathbb{Q}}$ such that $N\mathcal{O}_v \subset \mathfrak{m}_v\mathcal{O}_v \cap \pi_v\mathcal{O}_v$, where $\pi_v \in \mathcal{O}_v$ denotes a uniformizer. Let

$$R_{\alpha} := \left\{ r \in (\mathbb{Z}/N\mathbb{Z})^n \text{ such that } \beta_S(\tilde{r}) \in U_{\alpha} \times \prod_{v \in S' - S} \mathcal{O}_v \right\}.$$

For every $r \in R_{\alpha}$, consider the character

$$\chi_r : T[N] \longrightarrow \mathbb{C}^{\times}, \quad t \longmapsto e^{2\pi i r^T \tilde{t}},$$

where $\tilde{t} \in N^{-1}\mathbb{Z}^n/\mathbb{Z}^n$ represents the torsion point $t \in T[N]$. Therefore, we can describe

$$V_{\alpha} = \bigsqcup_{r \in R_{\alpha}} V_{\chi_r},$$

where V_{χ_r} was described in Section 4.4. Denote by $\Gamma_{V_\alpha} \subset \mathrm{SL}_n(\mathbb{Z})$ the stabilizer of V_α . Observe that $\rho(U_F(f)) \subset \Gamma_{V_\alpha}$. It then follows from Corollary 4.11 that

$${}_c\Theta^{\mathbf{D}}(c_{U_F(f)})(U_\alpha) = \int_{U_F(f) \backslash (F \otimes \mathbb{R})^1} j_\tau^* e^* \iota_W E_\psi(\phi_{V_\alpha, c}),$$

where $\phi_{V_\alpha, c}$ was defined at the end of Section 4.4 and we recall it is given by

$$\phi_{V_\alpha, c} = \sum_{r \in R_\alpha} \phi_{D_{\chi_r}, c}, \text{ with } \phi_{D_{\chi_r}, c}(v) = \frac{1}{N^n} \sum_{t \in \mathbf{T}[N]} \chi_r(t)^{-1} \phi_c(v - \tilde{t}).$$

We can now apply Proposition 4.14, to deduce that, if we denote

$$f(y) = \phi_{V_\alpha, c}((\beta^\vee)^{-1}(y)),$$

the previous period is equal to

$$\frac{\kappa_n(-i)^n}{\delta} Z_{\hat{f}}(0).$$

Finally, observe that the Fourier transform of f is given by

$$\hat{f}(x) = \delta \left(c^n 1_{\widehat{c\mathfrak{a}^{-1}}} - 1_{\widehat{\mathfrak{a}^{-1}(x)}} \right) 1_{U_\alpha}(x_S).$$

Therefore,

$$\begin{aligned} {}_c\Theta^{\mathbf{D}}(c_{U_F(f)})(U_\alpha) &= \kappa_n(-i)^n \left(c^n \sum_{\substack{y \in U_F(f) \backslash c\mathfrak{a}^{-1} \\ y_S \in U_\alpha}} \frac{\mathrm{sgn}(Ny)}{|Ny|^s} - \sum_{\substack{y \in U_F(f) \backslash \mathfrak{a}^{-1} \\ y_S \in U_\alpha}} \frac{\mathrm{sgn}(Ny)}{|Ny|^s} \right) \\ &= -(-1)^{n(n+1)/2} L_{S, c}(H_{\mathfrak{m}}/F, [(\tilde{\alpha})\mathfrak{a}]_{\mathfrak{m}}, 0). \end{aligned}$$

□

5. GROSS–STARK UNITS AND GROSS TOWER OF FIELDS CONJECTURE

Let F be a totally real field of degree n . Let $f \in \mathbb{Z}$ be a positive integer, and let H be the narrow ray class field of F of modulus $\mathfrak{f} = f\mathcal{O}$. We further assume that c is such that the only root of unity in H which is $1 \pmod{c\mathcal{O}_H}$ is 1. We suppose that there exists p a rational prime that is inert in F and $p \equiv 1 \pmod{f}$ and let $\mathfrak{p} := p\mathcal{O}$.

In this section, we use the class ${}_c\Theta^{\mathbf{D}}$ to produce a p -adic analytic formula for Gross–Stark units in H . Then, we prove this formula using the Gross tower of fields conjecture, proven in [DK24]. The strategy we follow is similar to the one of [Dup].

5.1. Gross–Stark units and tower of fields conjecture. In this section, we introduce Gross–Stark units and state the Gross’ tower of fields conjecture.

We suppose that there exists p a rational prime that is inert in F and $p \equiv 1 \pmod{f}$. Consider the following subgroup of p -units in H

$$\mathcal{O}_H[1/p]_-^\times := \{x \in H^\times \mid |x|_{\mathfrak{Q}} = 1 \ \forall \mathfrak{Q} \nmid \mathfrak{p}\},$$

where $\mathfrak{Q}$ runs over all archimedean and nonarchimedean places of H not dividing $\mathfrak{p}$. The embedding $\bar{\mathbb{Q}} \subset \bar{\mathbb{Q}}_p$ determines a prime $\mathfrak{P}$ of H above p that we fix here and from now on.

Let S be a finite set of places of F containing $\mathfrak{p}$, all the primes that divide f , and all the archimedean primes.

Theorem 5.1. *There exists a unique $u_S \in \mathcal{O}_H[1/p]_-^\times$ such that $u_S \equiv 1 \pmod{c\mathcal{O}_H[1/p]}$ and for every ideal $\mathfrak{a}$ prime to $S \cup \{c\}$*

$$\mathrm{ord}_{\mathfrak{p}}(u_S^{\sigma_{\mathfrak{a}}}) = L_{R,c}([\mathfrak{a}]_f, 0).$$

It will be helpful to relate the Gross–Stark units u_S for different choices of S .

Lemma 5.2. *Consider the same notation as above. Let $\mathfrak{q}$ be a prime disjoint from S . We have*

$$u_{S \cup \{\mathfrak{q}\}} = u_S / u_S^{\sigma_{\mathfrak{q}}^{-1}}.$$

Proof. For every $\sigma_{\mathfrak{a}} \in \mathrm{Gal}(H/F)$, we have

$$\mathrm{ord}_{\mathfrak{p}}(u_S^{\sigma_{\mathfrak{a}}} / u_S^{\sigma_{\mathfrak{a}}\sigma_{\mathfrak{q}}^{-1}}) = L_{R,c}([\mathfrak{a}]_f, 0) - L_{R,c}([\mathfrak{a}\mathfrak{q}^{-1}]_f, 0) = L_{R \cup \{\mathfrak{q}\},c}([\mathfrak{a}]_f, 0) = \mathrm{ord}_{\mathfrak{p}}(u_{S \cup \{\mathfrak{q}\}}^{\sigma_{\mathfrak{a}}}).$$

This implies the result by the unicity asserted above. $\square$

We now state the Gross’ tower of fields conjecture, proven by Dasgupta and Kakde. Let K be a finite abelian extension of F , containing H , unramified outside S . Consider

$$\mathrm{rec}_{\mathfrak{p}}: F_{\mathfrak{p}}^\times \longrightarrow \mathbb{A}_F^\times \longrightarrow \mathrm{Gal}(K/F).$$

Observe that since $\mathfrak{p}$ splits completely in H , we have $H \subset H_{\mathfrak{p}} \simeq F_{\mathfrak{p}}$. In particular, we can consider $\mathrm{rec}_{\mathfrak{p}}(u_S^{\sigma_{\mathfrak{a}}})$ and the image will belong to $\mathrm{Gal}(K/H)$.

Theorem 5.3. *Let $u_S \in \mathcal{O}_H[1/p]_-^\times$ be as above and let K be as above. For every $\sigma \in \mathrm{Gal}(H/F)$ we have*

$$\mathrm{rec}_{\mathfrak{p}}(u_S^{\sigma_{\mathfrak{a}}}) = \prod_{\tau \in \mathrm{Gal}(K/H)} \tau^{L_{S,c}(K/F, \tilde{\sigma}\tau^{-1}, 0)} \in \mathrm{Gal}(K/F),$$

where $\tilde{\sigma}$ is a lift of σ to $\mathrm{Gal}(K/F)$.

We will apply this theorem in the following setting. Let $\mathfrak{m}$ be an integral ideal of F such that $\mathfrak{f} \mid \mathfrak{m}$, and $\mathfrak{m}$ is supported in the primes in S . Let $K = H_{\mathfrak{m}}$ be the narrow ray class field of modulus $\mathfrak{m}$, which is an abelian extension of F satisfying $H_{\mathfrak{m}} \supset H$. By class field theory, we have an isomorphism

$$\mathrm{rec}_S: \mathcal{O}_{\mathfrak{p}}^\times \times \mathcal{U}/U_F(f)U_{\mathfrak{m}} \xrightarrow{\sim} \mathrm{Gal}(K/H).$$

Observe that we have the identifications

$$\mathcal{O}_{\mathfrak{p}}^\times \times \mathcal{U}/U_F(f)U_{\mathfrak{m}} \simeq H_{\mathfrak{m},[\mathfrak{a}]_{\mathfrak{f}}}, \quad \alpha \longmapsto [(\alpha)\mathfrak{a}]_{\mathfrak{m}},$$

where α is chosen to be totally positive. Let $U_F[1/p](f)$ denote the p -units in F that are congruent to 1 modulo F . Moreover, inclusion induces an isomorphism

$$\mathcal{O}_{\mathfrak{p}}^\times \times \mathcal{U}/U_F(f)U_{\mathfrak{m}} \xrightarrow{\sim} F_{\mathfrak{p}}^\times \times \mathcal{U}/U_F[1/p](f)U_{\mathfrak{m}}.$$

Then, by applying $\mathrm{rec}_{K/H}^{-1}$ on both sides of the last equation, we get the following equality in $F_{\mathfrak{p}}^\times \times \mathcal{U}/U_F[1/p](f)U_{\mathfrak{m}}$

$$(u_S^{\sigma_{\mathfrak{a}}}, 1, \dots, 1) = \prod_{\alpha \in \mathcal{O}_{\mathfrak{p}}^\times \times \mathcal{U}/U_F(f)U_{\mathfrak{m}}} \alpha^{L_{S,c}(K/F, \sigma_{\mathfrak{a}}\mathrm{rec}_S(\alpha^{-1}), 0)}.$$

Since α is prime to S , we have $\sigma_{(\alpha)} = \text{rec}_S(\alpha)^{-1}$. From there, the previous equality can be written as

$$(u_S^{\sigma_{\mathfrak{a}}}, 1, \dots, 1) = \prod_{\alpha \in \mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U}/U_F(f)U_{\mathfrak{m}}} \alpha^{L_{S,c}(K/F, \sigma_{\mathfrak{a}}\sigma_{(\alpha)}, 0)} = \prod_{\alpha \in \mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U}/U_F(f)U_{\mathfrak{m}}} \alpha^{L_{S,c}([\mathfrak{a}(\alpha)]_{\mathfrak{m}}, 0)}. \quad (7)$$

5.2. Gross tower of fields conjecture in terms of the measure valued class. In this section, we express the previous Gross tower of fields conjecture in using the cocycle ${}_c\Theta^{\mathbf{D}}$. Let

$$\mathbb{X}_{S_{\mathbb{Q}}} := \prod_{q \in S_{\mathbb{Q}}} \mathbb{Z}_q^n$$

and consider the pushforward of measures, which is $\text{GL}_n(\mathbb{Z}[1/S_{\mathbb{Q}}])$ -equivariant

$$\mathbf{D}(\mathbb{X}_0, \mathbb{Z})^{(0_c)}/\delta_0 \longrightarrow \mathbf{D}(\mathbb{X}_{S_{\mathbb{Q}}})^{(0)}/\delta_0,$$

where the superscript (0) on the right-hand side denotes invariant under multiplication by every integer supported in the primes in $S_{\mathbb{Q}}$. Let

$${}_c\Theta_S^{\mathbf{D}} \in H^{n-1}(\text{GL}_n^+(\mathbb{Z}[1/S_{\mathbb{Q}}]), \mathbf{D}(\mathbb{X}_{S_{\mathbb{Q}}})^{(0)}/\delta_0)$$

be the image of ${}_c\Theta^{\mathbf{D}}$ by the map above.

Recall that $\mathfrak{a}$ is a fractional ideal of F prime to $S \cup \{c\}$. Let $\tau \in F^n$ be a $\mathbb{Z}$ -basis of a fractional ideal $\mathfrak{a}^{-1}$, such that the isomorphism

$$\beta_{\tau}: \mathbb{Z}^n \xrightarrow{\sim} \mathfrak{a}^{-1}, \quad x \longmapsto \tau^t \cdot x$$

satisfies that $\beta_{\tau}(e_1) \equiv 1 \pmod{f_v \mathfrak{a}_v^{-1}}$ for every finite place $v \mid f$. This bijection induces an embedding

$$F \hookrightarrow M_n(\mathbb{Q}), \quad \alpha \longmapsto A_{\alpha},$$

and let

$$\rho_{\tau}: U_F(f) \hookrightarrow \text{SL}_n(\mathbb{Z}), \quad u \longmapsto A_u^{-T}.$$

Using the isomorphism β_{τ} , we can view

$$\mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U} \times \prod_{v \in S' - S} \mathcal{O}_v \xrightarrow{\beta_{\tau}^{-1}} \mathbb{X}'_{S_{\mathbb{Q}}}.$$

We can consider the map of measures

$$\varphi^{\text{tame}}: \mathbf{D}(\mathbb{X}_{S_{\mathbb{Q}}})/\delta_0 \longrightarrow \mathbf{D}(\mathbb{X}'_{S_{\mathbb{Q}}})^{(0)} \xrightarrow{\beta_{\tau}^{-1}} \mathbf{D}_0(\mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U} \times \prod_{v \in S' - S} \mathcal{O}_v) \longrightarrow \mathbf{D}_0(\mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U}),$$

where the first two maps are given by restriction, and the last one is pushforward. Define also

$$\varphi: \mathbf{D}(\mathbb{X}_{S_{\mathbb{Q}}})^{(0)} \xrightarrow{\varphi^{\text{tame}}} \mathbf{D}_0(\mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U}) \longrightarrow \mathbf{D}_0(\mathcal{O}_{\mathfrak{p}}^{\times})$$

by composing φ^{tame} with the pushforward induced from the projection $\mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U} \twoheadrightarrow \mathcal{O}_{\mathfrak{p}}^{\times}$. The following lemma justifies that the previous maps are well-defined.

Lemma 5.4. *The map φ^{tame} lands in the space of measures on $\mathcal{O}_{\mathfrak{p}}^{\times}$ of total mass zero. Similarly, φ lands in the space of measures on $\mathcal{O}_{\mathfrak{p}}^{\times}$ of total mass zero. Moreover, both maps are $U_F(f)$ -equivariant, where $U_F(f)$ acts on $\mathbf{D}(\mathbb{X}_{S_{\mathbb{Q}}})$ via ρ_{τ} , and on $\mathbf{D}(\mathcal{O}_{\mathfrak{p}}^{\times})$ via multiplication.*

Definition 5.5. Let $\mu_{S,\tau}^{\text{tame}} \in H^{n-1}(U_F(f), \mathbf{D}_0(\mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U}))$ be the image of ${}_c\Theta_S^{\mathbf{D}}$ by the map φ^{tame} given above. Let $\mu_{S,\tau} \in H^{n-1}(U_F(f), \mathbf{D}_0(\mathcal{O}_{\mathfrak{p}}^{\times}))$ be the image of ${}_c\Theta_S^{\mathbf{D}}$ via φ .

Finally, we can rewrite the quantity appearing in the Gross tower of fields conjecture as follows. Consider now the $U_F(f)$ -equivariant morphism

$$\varphi_{\mathfrak{m}}: \mathbf{D}(\mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U}) \longrightarrow (\mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U}) / U_F(f)U_{\mathfrak{m}}, \lambda \longmapsto \prod_{\alpha \in \mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U} / U_F(f)U_{\mathfrak{m}}} \alpha^{\lambda(U_{\alpha})}. \quad (8)$$

Theorem 5.6. *We have*

$$\varphi_{\mathfrak{m}}(\mu_{S,\tau}^{\text{tame}}(c_{U_F(f)})) = (u_S^{\sigma_{\mathfrak{a}}}, 1) \text{ in } (F_{\mathfrak{p}}^{\times} \times \mathcal{U}) / U_F[1/p](f)U_{\mathfrak{m}}.$$

5.3. Formula for Gross–Stark units. In this section, we propose a formula for the Gross–Stark units $u_S^{\sigma_{\mathfrak{a}}}$ in the narrow ray class field of F of modulus $\mathfrak{f} = f\mathcal{O}_F$. Then, we show some properties about the formula, including that it only depends on the ideal class of $\mathfrak{a}$ in $G_{\mathfrak{f}}$, and study the dependencies on the set S .

Recall the cocycles valued on measures $\mu_{S,\tau}^{\text{tame}}$ and $\mu_{S,\tau}$ defined above. Observe that the maps

$$\mathbf{D}_0(\mathcal{O}_{\mathfrak{p}}^{\times}, \mathbb{Z}) \longrightarrow \mathcal{O}_{\mathfrak{p}}^{\times}, \lambda \longmapsto \oint_{\mathcal{O}_{\mathfrak{p}}^{\times}} x d\lambda$$

$$\mathbf{D}_0(\mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U}) \longrightarrow \mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U}, \lambda \longmapsto \oint_{\mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U}} x d\lambda$$

are $U_F(f)$ -equivariant, where the action on $U_F(f)$ is induced by multiplication on the measures, and it is trivial on the target.

Let $c_{U_F(f)} \in H_{n-1}(U_F(f), \mathbb{Z}) \simeq \mathbb{Z}$ be a fundamental class corresponding to a positively oriented basis of $U_F(f)$. We can then consider the following cap products.

Definition 5.7. Define

$$v_S(\tau) := p^{L_{R,c}([\mathfrak{a}]_{\mathfrak{f}}, 0)} \oint_{\mathcal{O}_{\mathfrak{p}}^{\times}} x d\mu_{S,\tau,f}(c_{U_F(f)}) \in F_{\mathfrak{p}}^{\times},$$

$$v_S^{\text{tame}}(\tau) := p^{L_{R,c}([\mathfrak{a}]_{\mathfrak{f}}, 0)} \oint_{\mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U}} (x, \dots, x) d\mu_{S,\tau,f}^{\text{tame}}(c_{U_F(f)}) \in F_{\mathfrak{p}}^{\times} \times \mathcal{U}.$$

Lemma 5.8. *Suppose that $\mathfrak{a}$ is prime to $S_{\mathbb{Q}}$. The invariant $v_S(\tau)$ depends only on the ideal class of $\mathfrak{a}$ in $G_{\mathfrak{f}}$.*

Proof. We first show that $v_S(\tau)$ only depends on the ideal $\mathfrak{a}$. Namely, if $\tau_1, \tau_2 \in F^n$ are two positively oriented basis of $\mathfrak{a}^{-1}$ such that $\beta_{\tau_1}(e_1) \equiv \beta_{\tau_2}(e_1) \equiv 1 \pmod{f_v \mathfrak{a}_v^{-1}}$ for every $v \mid f$, then $v_S(\tau_1) = v_S(\tau_2)$. For that, note that in such case, $\tau_1 = \gamma \tau_2$ for some $\gamma \in \text{SL}_n(\mathbb{Z})$ such that $\gamma^t \in \Gamma_1(f)$. We then have a commutative diagram

$$\begin{array}{ccc}
\prod_{q \in S_{\mathbb{Q}}} \mathbb{Z}_q^n & \xrightarrow{x \mapsto \gamma^t x} & \prod_{q \in S_{\mathbb{Q}}} \mathbb{Z}_q^n \\
\downarrow x \mapsto \tau_1^t \cdot x & & \downarrow x \mapsto \tau_2^t \cdot x \\
\prod_{q \in S_{\mathbb{Q}}} \prod_{q|q} \mathfrak{a}_q^{-1} & \xrightarrow{\text{Id}} & \prod_{q \in S_{\mathbb{Q}}} \prod_{q|q} \mathfrak{a}_q^{-1} \\
\uparrow & & \uparrow \\
\mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U} \times \prod_{q \in S' - S} \mathcal{O}_q & \xrightarrow{\text{Id}} & \mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U} \times \prod_{q \in S' - S} \mathcal{O}_q
\end{array}$$

Denote by V_{τ_i} the open subset of $\prod_{q \in S_{\mathbb{Q}} - p} \mathbb{Z}_q^n$ corresponding to $\mathcal{U} \times \prod_{q \in S' - S} \mathcal{O}_q$ via the bijection β_{τ_i} . It follows from the commutative diagram above that $V_{\tau_2} = \gamma^t V_{\tau_1}$. Also, if $U_{\tau_i} = \rho_{\tau_i}(U_F(f))$ and $c_{\tau_i} \in H_{n-1}(U_{\tau_i}, \mathbb{Z})$ is the image of $c_{U_F(f)}$ via ρ_{τ_i} , we have note that $c_{U_{\tau_2}} = \gamma^{-1} c_{U_{\tau_1}} \gamma$. Then, from the definition of $v_S(\tau_i)$, we have

$$\frac{v_S(\tau_i)}{p^{L_{R,c}([\mathfrak{a}]_f, 0)}} = \lim_{U_x} \prod \tau_i^t \cdot x^{\Theta_S(c_{U_{\tau_1}})(U_x \times V_{\tau_i})}.$$

Thus,

$$\begin{aligned}
\frac{v_S(\tau_2)}{p^{L_{R,c}([\mathfrak{a}]_f, 0)}} &= \lim_{U_x} \prod (\tau_2^t \cdot x)^{\Theta_S(c_{U_{\tau_2}})(U_x \times V_2)} \\
&= \lim_{U_x} \prod (\tau_1^t (\gamma^t)^{-1} x)^{\Theta_S(c_{U_{\tau_2}})(U_x \times \gamma^t V_{\tau_1})} \\
&= \lim_{U_x} \prod (\tau_1^t \cdot x)^{\Theta_S(c_{U_{\tau_2}})(\gamma^t(U_x \times V_{\tau_1}))} \\
&= \lim_{U_x} \prod (\tau_1^t \cdot x)^{\Theta_S(\gamma^{-1} c_{U_{\tau_1}} \gamma)(\gamma^t(U_x \times V_{\tau_1}))} \\
&= \frac{v_S(\tau_1)}{p^{L_{R,c}([\mathfrak{a}]_f, 0)}},
\end{aligned}$$

where in the last equality, we used that the action of $\gamma \in \text{SL}_n(\mathbb{Z})$ on $H^{n-1}(\text{SL}_n(\mathbb{Z}), \mathbf{D}(\mathbb{X}'_{S_{\mathbb{Q}}}))$ induced by conjugation by γ on $\text{SL}_n(\mathbb{Z})$, and translation by γ^t on the measures, is the identity on cohomology.

We now see that $v_S(\tau)$ depends only on the ideal class of $\mathfrak{a}$ in $G_{\mathfrak{f}}$. Suppose that $\mathfrak{a}_1$ and $\mathfrak{a}_2$ are prime to $S_{\mathbb{Q}}$ and in the same class in $G_{\mathfrak{f}}$. Then, there exists $\lambda \in F$ totally positive such that $\mathfrak{a}_2 = (\lambda)\mathfrak{a}_1$ and $\lambda \equiv 1 \pmod{f_v \mathcal{O}_v}$ for every $v \mid f$. Choose $\tau \in F^n$ a $\mathbb{Z}$ -basis of $\mathfrak{a}_1^{-1}$ such that $\beta_{\tau}(e_1) \equiv 1 \pmod{f}$. Then, $\lambda^{-1}\tau \in F^n$ is a $\mathbb{Z}$ -basis of $\mathfrak{a}_2^{-1}$. Observe that since $\mathfrak{a}_1$ and $\mathfrak{a}_2$ are prime to $S_{\mathbb{Q}}$, it follows that so is (λ) . From there, we deduce that multiplication by λ stabilizes $\mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U} \times \prod_{q|q} \mathcal{O}_q$ and therefore that $V_{\tau} = V_{\lambda^{-1}\tau}$. Finally, also note that $U_{\tau} = U_{\lambda^{-1}\tau}$. From there,

$$v_S(\lambda\tau) = p^{L_{R,c}([\mathfrak{a}]_f, 0)} \lim_{U_x} \prod (\lambda\tau^t \cdot x)^{\Theta_S(c_{U_{\tau}})(U_x \times V_{\tau})} = \lambda^{\Theta_S(c_{U_{\tau}})((\mathbb{Z}_p^n)' \times V_{\tau})} v_S(\tau)$$

and the result follows as $\Theta_S(c_{U_{\tau}})$ evaluated on $(\mathbb{Z}_p^n)' \times V_{\tau}$ is equal to 0 by the total mass zero property. $\square$

In view of the previous lemma, we can make the following definition.

Definition 5.9. For $\mathfrak{a}$ an ideal prime to $S_{\mathbb{Q}}$, denote $v_S([\mathfrak{a}]) = v_S(\tau)$, for $\tau \in F^n$ a $\mathbb{Z}$ -basis of $\mathfrak{a}^{-1}$ such that $\beta(e_1) \equiv 1 \pmod{f_v \mathfrak{a}_v^{-1}}$ for every $v \mid f$.

The following two lemmas show that the quantities $v_S([\mathfrak{a}])$ have similar transformation properties as the Gross–Stark units.

Lemma 5.10. *Suppose that $\mathfrak{r}$ is a prime ideal of F that is coprime to $S_{\mathbb{Q}}$. Suppose that $\mathfrak{a}$ is prime to $S_{\mathbb{Q}}$ and to $N\mathfrak{r}$. We have*

$$v_{S \cup \{\mathfrak{r}\}}([\mathfrak{a}]) = v_S([\mathfrak{a}]) / v_S([\mathfrak{a}\mathfrak{r}^{-1}]).$$

Proof. We will first write the formulas for each of the terms appearing in the formula. Let r be the rational prime below $\mathfrak{r}_1$, and denote $S(\mathfrak{r}) = S \cup \{\mathfrak{r}\}$ and consider the corresponding $S_{\mathbb{Q}}(r)$ and $S'(r)$.

(1) Formula for $v_S([\mathfrak{a}])$. Choose $\tau_1 \in F^n$ a $\mathbb{Z}$ -basis of $\mathfrak{a}^{-1}$ such that

$$\beta_{\tau_1}: \mathbb{Z}^n \xrightarrow{\sim} \mathfrak{a}^{-1}, \quad x \mapsto \tau_1^t \cdot x$$

satisfies $\beta_{\tau_1}(e_1) \equiv 1 \pmod{f_v \mathcal{O}_v}$ for every $v \mid f$. Following the notation of the proof of the previous lemma, we can write

$$\begin{aligned} v_S([\mathfrak{a}]) &= \lim_{U_x} \prod (\tau_1^t \cdot x)^{\Theta_S(c_{U_{\tau_1}})(U_x \times V_{\tau_1})} \\ &= \lim_{U_x} \prod (\tau_1^t \cdot x)^{\Theta_{S(\mathfrak{r})}(c_{U_{\tau_1}})(U_x \times V_{\tau_1} \times \mathbb{Z}_r^n)}, \end{aligned}$$

where in the last equality, we used the definition of Θ_S and $\Theta_{S \cup \{\mathfrak{r}\}}$ as the pushforward of ${}_c\Theta^{\mathbf{D}}$.

(2) Formula for $v_S([\mathfrak{a}\mathfrak{r}^{-1}])$. Choose $\tau_2 \in F^n$ a positively oriented $\mathbb{Z}$ -basis of $\mathfrak{a}^{-1}\mathfrak{r}$, i.e.

$$\beta_{\tau_2}: \mathbb{Z}^n \xrightarrow{\sim} \mathfrak{a}^{-1}\mathfrak{r}, \quad x \mapsto \tau_2^t \cdot x$$

such that $\beta_{\tau_2}(e_1) \equiv 1 \pmod{f_v \mathfrak{a}_v^{-1} \mathfrak{r}_v}$ for every $v \mid f$. Note that β_{τ_2} induces an isomorphism

$$\prod_{q \in S_{\mathbb{Q}}(r)} \mathbb{Z}_q^n \xrightarrow{\sim} \prod_{q \in S_{\mathbb{Q}}} \prod_{q \mid q} \mathfrak{a}_q^{-1} \times \mathfrak{r} \mathfrak{a}_{\mathfrak{r}}^{-1} \times \prod_{\substack{\mathfrak{r}' \mid r \\ \mathfrak{r}' \neq \mathfrak{r}}} \mathfrak{a}_{\mathfrak{r}'}^{-1}$$

Let $V_{\tau_2} \subset \prod_{q \in S_{\mathbb{Q}} - \{p\}} \mathbb{Z}_q^n$ be such that

$$\beta_{\tau_2}(V_{\tau_2}) = \mathcal{U} \times \prod_{q \in S' - S} \mathcal{O}_q.$$

Similarly as before, the formula $v_S([\mathfrak{a}\mathfrak{r}^{-1}])$ can be written as

$$\begin{aligned} v_S([\mathfrak{a}\mathfrak{r}^{-1}]) &= \lim_{U_x} \prod (\tau_2^t \cdot x)^{\Theta_S(c_{U_{\tau_2}})(U_x \times V_{\tau_2})} \\ &= \lim_{U_x} \prod (\tau_2^t \cdot x)^{\Theta_{S(\mathfrak{r})}(c_{U_{\tau_2}})(U_x \times V_{\tau_2} \times \mathbb{Z}_r^n)}. \end{aligned}$$

(3) Formula for $v_{S(\mathfrak{v})}([\mathfrak{a}])$. Note that β_{τ_1} induces the isomorphism

$$\prod_{q \in S_{\mathbb{Q}}(r)} \mathbb{Z}_q^n \xrightarrow{\sim} \prod_{q \in S_{\mathbb{Q}}(r)} \prod_{q|q} \mathfrak{a}_q^{-1}.$$

Let $M \in \mathrm{GL}_n(\mathbb{Q})$ be such that $M\tau_1 = \tau_2$. Note that in fact $M \in \mathrm{GL}_n^+(\mathbb{Z}[1/r])$. Then, β_{τ_1} induces a bijection

$$M^t((\mathbb{Z}_p^n)' \times V_{\tau_2} \times \mathbb{Z}_r^n) \xrightarrow{\sim} \mathcal{O}_{\mathfrak{p}}^\times \times \mathcal{U} \times \prod_{q \in S' - S} \mathcal{O}_q \times \mathfrak{r}\mathcal{O}_{\mathfrak{r}} \times \prod_{\substack{\mathfrak{r}'|r \\ \mathfrak{r}' \neq \mathfrak{r}}} \mathcal{O}_{\mathfrak{r}'}.$$

Moreover $V_{\tau_1} = M^t V_{\tau_2}$. Therefore,

$$v_{S(\mathfrak{r}_1)}([\mathfrak{a}]) = \lim_{U_x} \prod (\tau_1^t \cdot x)^{\Theta_{S(\mathfrak{v})}(c_{U_{\tau_1}})(U_x \times V_{\tau_1} \times (\mathbb{Z}_r^n - M^t \mathbb{Z}_r^n))}.$$

We now wish to rewrite $v_{S(\mathfrak{r}_1)}([\mathfrak{a}])$ in terms of $v_S([\mathfrak{a}])$ and $v_S([\mathfrak{a}\mathfrak{r}_1^{-1}])$. Observe

$$v_{S(\mathfrak{r}_1)}([\mathfrak{a}]) = \frac{\lim_{U_x} \prod (\tau_1^t \cdot x)^{\Theta_{S(\mathfrak{v})}(c_{U_{\tau_1}})(U_x \times V_{\tau_1} \times \mathbb{Z}_r^n)}}{\lim_{U_x} \prod (\tau_1^t \cdot x)^{\Theta_{S(\mathfrak{v})}(c_{U_{\tau_1}})(U_x \times V_{\tau_1} \times M^t \mathbb{Z}_r^n)}}.$$

Observe that the numerator is equal to $v_S([\mathfrak{a}])$. We now focus on the denominator. Since $M\tau_1 = \tau_2$, it follows $c_{U_{\tau_2}} = M c_{U_{\tau_1}} M^{-1}$. From there,

$$\begin{aligned} \lim_{U_x} \prod (\tau_1^t \cdot x)^{\Theta_{S(\mathfrak{v})}(c_{U_{\tau_1}})(U_x \times V_{\tau_1} \times M^t \mathbb{Z}_r^n)} &= \lim_{U_x} \prod (\tau_2^t \cdot (M^t)^{-1} x)^{\Theta_{S(\mathfrak{v})}(c_{U_{\tau_1}})(U_x \times M^t V_{\tau_2} \times M^t \mathbb{Z}_r^n)} \\ &= \lim_{U_x} \prod (\tau_2^t \cdot x)^{\Theta_{S(\mathfrak{v})}(c_{U_{\tau_1}})(M^t U_x \times M^t V_{\tau_2} \times M^t \mathbb{Z}_r^n)} \\ &= \lim_{U_x} \prod (\tau_2^t \cdot x)^{\Theta_{S(\mathfrak{v})}(M^{-1} c_{U_{\tau_2}} M)(M^t U_x \times M^t V_{\tau_2} \times M^t \mathbb{Z}_r^n)} \\ &= \lim_{U_x} \prod (\tau_2^t \cdot x)^{\Theta_{S(\mathfrak{v})}(c_{U_{\tau_2}})(U_x \times V_{\tau_2} \times \mathbb{Z}_r^n)} \\ &= v_S([\mathfrak{a}\mathfrak{r}_1^{-1}]), \end{aligned}$$

where in the second to last equality we used that $\Theta_{S(\mathfrak{v})}$ is a cocycle for $\mathrm{GL}_n^+(\mathbb{Z}[1/S_{\mathbb{Q}}(r)])$, and therefore the natural conjugation action of M in $\mathrm{GL}_n^+(\mathbb{Z}[1/S_{\mathbb{Q}}(r)])$ and translation on $\mathbf{D}(\mathbb{X}'_{S_{\mathbb{Q}}(r)})^{(0)}$ fixes the class of $\Theta_{S(\mathfrak{v})}$. $\square$

Lemma 5.11. *Suppose that $\mathfrak{r}$ is a prime ideal of F such that $\sigma_{\mathfrak{r}} \in \mathrm{Gal}(H_{\mathfrak{f}}/F)$ is equal to complex conjugation. Then,*

$$v_S([\mathfrak{a}\mathfrak{r}]) = v_S([\mathfrak{a}])^{-1}.$$

5.4. Tame refinement. Recall the definition of $\mu_{S,\tau}^{\mathrm{tame}}$ and of $v_S^{\mathrm{tame}}(\tau)$. In this section, we calculate v_S^{tame} in terms of $v_S([\mathfrak{a}])$ and verify that the image of v_S^{tame} in $F_{\mathfrak{p}}^\times \times \mathcal{U}/\overline{U_F(f)}$ is equal to a Gross–Stark unit.

Proposition 5.12. *Suppose that $S_{\mathrm{fin}} \supsetneq \{\mathfrak{p}\}$. We have*

$$v_S^{\mathrm{tame}}(\tau) = (v_S(\tau), 1, \dots, 1) \in F_{\mathfrak{p}}^\times \times \mathcal{U}.$$

Proof. The fact that the first component of $v_S^{\text{tame}}(\tau)$ is equal to $v_S(\tau) = v_S([\mathfrak{a}])$ follows because $\mu_{S,\tau}$ is the pushforward of $\mu_{S,\tau}^{\text{tame}}$ induced by the projection map $\mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U} \rightarrow \mathcal{O}_{\mathfrak{p}}^{\times}$ (see Definition 5.5). To study the other components consider the inclusion

$$\beta_{\tau}^{-1}: \mathcal{O}_{\mathfrak{p}} \times \mathcal{U} \times \prod_{v \in S' - S} \mathcal{O}_v \hookrightarrow \prod_{q \in S_{\mathbb{Q}}} \mathbb{Z}_q^n,$$

and note that the image does not contain 0, as $R \neq \emptyset$. Thus, we can consider the maps of measures

$$\mathbf{D}(\mathbb{X}_S)^{(0)}/\delta_0 \longrightarrow \mathbf{D}\left(\mathcal{O}_{\mathfrak{p}} \times \mathcal{U} \times \prod_{v \in S' - S} \mathcal{O}_v\right)^{(0)} \longrightarrow \mathbf{D}(\mathcal{O}_{\mathfrak{p}} \times \mathcal{U})^{(0)},$$

where the first map is given by restriction (via the map β_{τ}^{-1} above), and the second one via pushforward. Observe that the previous map is $U_F(f)$ -equivariant. Let

$$\tilde{\mu}_{S,\tau} \in H^{n-1}(U_F(f), \mathbf{D}(\mathcal{O}_{\mathfrak{p}} \times \mathcal{U})^{(0)}),$$

be the image of Θ_S by this map. Fix a representative of $\tilde{\mu}_{S,\tau}$, that we will denote by the same symbol. Fix $\mathfrak{q}_0 \in R$ and define the cocycles

$$\nu_0, \nu_1 \in Z^{n-1}(U_F(f), \mathbf{D}(U_{\mathfrak{q}_0,f}, \mathbb{Z}))$$

as follows: for $\underline{\gamma} \in U_F(f)^n$, and $U \subset U_{\mathfrak{q}_0,f}$ let

$$\nu_0(\underline{\gamma})(U) = \tilde{\mu}_{S,\tau}(\underline{\gamma})(\mathcal{O}_{\mathfrak{p}} \times U \times \prod_{\mathfrak{q} \in R - \{\mathfrak{q}_0\}} U_{\mathfrak{q},f}),$$

$$\nu_1(\underline{\gamma})(U) = \tilde{\mu}_{S,\tau,f}(\underline{\gamma})(p\mathcal{O}_{\mathfrak{p}} \times U \times \prod_{\mathfrak{q} \in R - \{\mathfrak{q}_0\}} U_{\mathfrak{q},f}).$$

Since $\mathcal{O}_{\mathfrak{p}}^{\times} = \mathcal{O}_{\mathfrak{p}} - p\mathcal{O}_{\mathfrak{p}}$, we can write

$$\text{pr}_{\mathfrak{q}_0}(v_S^{\text{tame}}(\tau)) = p^{L_{R,c}([\mathfrak{a}]_f, 0)} \left(\int_{\mathcal{O}_{\mathfrak{q}_0}^{\times}} x d\nu_0(c_{U_F(f)}) \right) \left(\int_{\mathcal{O}_{\mathfrak{q}_0}^{\times}} x d\nu_1(c_{U_F(f)}) \right)^{-1}.$$

On the other hand, from the fact that $\tilde{\mu}_{S,\tau}$ takes values in p -invariant measures, we have $\nu_0(\underline{\gamma})(U) = \nu_1(\underline{\gamma})(pU)$ for every $\underline{\gamma}$ and U . From there, it follows that

$$\text{pr}_{\mathfrak{q}}(v_S^{\text{tame}}) = p^{L_{R,c}([\mathfrak{a}]_f, 0)} \left(\int_{\mathcal{O}_{\mathfrak{q}}^{\times}} p d\nu_0(c_{U_F(f)}) \right)^{-1} = p^{L_{R,c}([\mathfrak{a}]_f)} \mu_{S,\tau}(c_{U_F(f)})(\mathcal{O}_{\mathfrak{p}} \times \mathcal{U})^{-1}$$

and this is equal to 1, as the value of the class is a period of an Eisenstein series by Theorem 4.15. $\square$

Proposition 5.13. *We have*

$$v_S^{\text{tame}}(\tau) = (u_S^{\sigma_{\mathfrak{a}}}, 1, \dots, 1) \in F_{\mathfrak{p}}^{\times} \times \mathcal{U}/\overline{U_F(f)}.$$

Proof. The p -adic valuation is equal on both sides. Thus it is enough to prove that

$$p^{-L_{R,c}([\mathfrak{a}]_f, 0)} v_S^{\text{tame}}(\tau) = p^{-L_{R,c}([\mathfrak{a}]_f, 0)} (u_S^{\sigma_{\mathfrak{a}}}, 1) \in \mathcal{O}_{\mathfrak{p}}^{\times} \times \mathcal{U}/\overline{U_F(f)}.$$

Let $\mathfrak{m}$ be any ideal supported in the primes dividing S . To prove this last equality, it suffices to show it modulo $U_{\mathfrak{m}}$ for every such $\mathfrak{m}$. For a given $\mathfrak{m}$ as above, note that $v_S^{\text{tame}}(\tau)$ modulo

$\overline{U_F(f)}U_m$ is equal to $\varphi_m(\mu_{S,\tau}^{\text{tame}})$ given in Theorem 5.6, and the result follows from Theorem 5.6. $\square$

Let

$$D(\mathfrak{f}, \mathfrak{g}) := \{x \in F_{\mathfrak{p}}^{\times} \mid (x, 1) \in \overline{U_F(f)} \subset F_{\mathfrak{p}}^{\times} \times \mathcal{U}\}.$$

We then obtain the following corollary as a consequence Proposition 5.12 and Proposition 5.13.

Corollary 5.14. *We have*

$$v_S([\mathfrak{a}]) = u_S^{\sigma_{\mathfrak{a}}} \pmod{D(\mathfrak{f}, \mathfrak{g})}.$$

5.5. Proof of the formula for Gross–Stark units. We conclude this section by proving that $v_S([\mathfrak{a}]) = u_S^{\sigma_{\mathfrak{a}}}$ up to roots of unity in $F_{\mathfrak{p}}^{\times}$.

We begin with the following lemma from Dasgupta. Fix $\mathfrak{q}$ a prime ideal of F such that $\sigma_{\mathfrak{q}} \in \text{Gal}(H/F)$ is equal to a complex conjugation.

Lemma 5.15. *Let $m \geq 0$. There exist a finite set of prime ideals $\{\mathfrak{r}_1, \dots, \mathfrak{r}_s\}$ in the narrow ray class group of $\mathfrak{q}$ and such that*

- (1) *They are prime to $S_{\mathbb{Q}}$ and they are all of different residue characteristic.*
- (2) *The reduction of $D(\mathfrak{f}, \mathfrak{r}_1 \cdots \mathfrak{r}_s)$ modulo $\mathfrak{p}^m$ is contained in the image of the group W_0 of roots of unity inside $F_{\mathfrak{p}}^{\times}$.*

Theorem 5.16. *We have*

$$u_S^{\sigma_{\mathfrak{a}}} = v_S([\mathfrak{a}]).$$

Proof. Fix $m \geq 0$. Since W_0 is finite, it is enough to see that there exists $\zeta \in W_0$ such that

$$u_S^{\sigma_{\mathfrak{a}}} = \zeta v_S([\mathfrak{a}]) \pmod{\mathfrak{p}^m}.$$

In other words, it is enough to see that

$$(u_S/v_S([\mathfrak{a}])) = 1 \text{ in } \mathcal{O}_{\mathfrak{p}}^{\times}/W_0(1 + \mathfrak{p}^m \mathcal{O}_{\mathfrak{p}}).$$

Let $\mathfrak{r}_1, \dots, \mathfrak{r}_n$ be prime ideals satisfying the conditions of Lemma 5.15. Observe that

$$u_{S \cup \{\mathfrak{r}_1, \dots, \mathfrak{r}_s\}}^{\sigma_{\mathfrak{a}}} = (u_S^{\sigma_{\mathfrak{a}}})^{2^s}.$$

Similarly, by Lemma 5.10 and Lemma 5.11,

$$v_{S \cup \{\mathfrak{r}_1, \dots, \mathfrak{r}_s\}}([\mathfrak{a}]) = v_S([\mathfrak{a}])^{2^s}.$$

Now, it follows from Corollary 5.14 that

$$(u_S^{\sigma_{\mathfrak{a}}}/v_S([\mathfrak{a}]))^{2^s} = 1 \pmod{D(\mathfrak{f}, \mathfrak{r}_1 \cdots \mathfrak{r}_s)}.$$

Reducing the equality modulo $\mathfrak{p}^m$, it implies that there exists $\zeta \in W_0$ such that

$$(u_S^{\sigma_{\mathfrak{a}}}/v_S([\mathfrak{a}]))^{2^s} = 1 \text{ in } \mathcal{O}_{\mathfrak{p}}^{\times}/W_0(1 + \mathfrak{p}^m \mathcal{O}_{\mathfrak{p}}).$$

But now, observe that $\mathcal{O}_{\mathfrak{p}}^{\times}/W_0(1 + \mathfrak{p}^m \mathcal{O}_{\mathfrak{p}})$ is a p -group. Since p is odd, it follows that the function $x \mapsto x^{2^s}$ is bijective. Thus, we must have $(u_S^{\sigma_{\mathfrak{a}}}/v_S([\mathfrak{a}])) = 1$ in $\mathcal{O}_{\mathfrak{p}}^{\times}/W_0(1 + \mathfrak{p} \mathcal{O}_{\mathfrak{p}})$ as desired. $\square$

6. RIGID ANALYTIC EISENSTEIN CLASSES

Let $\mathcal{X}_p$ be Drinfeld's p -adic symmetric domain, namely $\mathbb{X}_p := \mathbb{P}^{n-1}(\mathbb{C}_p) - \cup_{\alpha} H_{\alpha}$, where α runs over the union all $\mathbb{Q}_p$ -rational hyperplanes in $\mathbb{P}^{n-1}(\mathbb{C}_p)$, equipped with the structure of rigid analytic space. In this section, we introduce the multiplicative van der Put lift $\text{ST}^{\times}$, which is an $\text{GL}_n(\mathbb{Q}_p)$ -equivariant map from $\mathbf{D}_0(\mathbb{X}, \mathbb{Z})$ to $\mathcal{A}^{\times}/p^{\mathbb{Z}}$, where $\mathcal{A}^{\times}$ is the multiplicative group of invertible functions on $\mathcal{X}_p$. Then, we use the distribution valued cocycles of the previous sections to construct rigid analytic classes in $H^{n-1}(\Gamma_1^p(f), \mathcal{A}^{\times}/p^{\mathbb{Z}})$. Finally, we define the evaluation of these cocycles at totally real fields of degree n where p is inert and prove these values are Gross–Stark p -units in the narrow ray class field of F of modulus $f\mathcal{O}$.

6.1. The Van der Put lift on Drinfeld's upper half-space. We describe a transform from p -invariant measures on $\mathbb{Y}_p := \mathbb{Q}_p^n - \{0\}$ to rigid analytic functions on $\mathcal{X}_p$. Similar transforms were considered by Van de Put, Schneider–Teitelbaum, and Darmon–Vonk.

The group $\text{GL}_n(\mathbb{Q}_p)$ acts on $\mathbb{P}^{n-1}(\mathbb{C}_p)$ by matrix multiplication and the $\mathbb{Q}_p$ -rational hyperplanes are preserved by this action. From there, we define a left action of $\text{SL}_n(\mathbb{Q}_p)$ on the space of functions on $\mathcal{X}_p$ given as follows. If $g \in \text{SL}_n(\mathbb{Q}_p)$, f is a function on $\mathcal{X}_p$, and $\tau \in \mathcal{X}_p$

$$(g \cdot f)(\tau) := f(g^{-1}\tau).$$

Denote by $\mathbb{X}_p := \mathbb{Z}_p^n - p\mathbb{Z}_p^n$, let $\mathbb{X}'_p$ be the primitive vectors on $\mathbb{X}_p$ and let $\mathbb{Y}_p := \mathbb{Q}_p^n - \{0\}$. Consider the space of $\mathbb{Z}$ -valued measures on $\mathbb{Y}_p$, denoted $\mathbf{D}(\mathbb{Y}_p, \mathbb{Z})$, let $\mathbf{D}(\mathbb{Y}_p, \mathbb{Z})^{(0)}$ be the submodule of measures that are p -invariant and let $\mathbf{D}_0(\mathbb{Y}_p, \mathbb{Z})^{(0)}$ be the module of p -invariant measures λ such that $\lambda(\mathbb{X}'_p) = 0$.

Consider the left action of $\text{GL}_n(\mathbb{Q}_p)$ on $\mathbb{Y}_p$ by $g \cdot y = g^{-t}y$, where $g \in \text{GL}_n(\mathbb{Q}_p)$ and $y \in \mathbb{Y}_p$. This induces a left action of $\text{GL}_n(\mathbb{Q}_p)$ on functions on $\mathbb{Y}_p$, by $(g \cdot h)(y) = h(g^t y)$ and on $\mathbf{D}(\mathbb{Y}_p, \mathbb{Z})$ given as follows: for $\lambda \in \mathbf{D}(\mathbb{Y}_p, \mathbb{Z})$ and $U \subset \mathbb{Y}_p$ compact open

$$(g \cdot \lambda)(U) = \lambda(g^t U).$$

Denote by $\mathbf{D}(\mathbb{Y}_p, \mathbb{Z})^{(p)} \subset \mathbf{D}(\mathbb{Y}_p, \mathbb{Z})$ the submodule of measures that are p -invariant and let $\mathbf{D}_0(\mathbb{Y}_p, \mathbb{Z})^{(0)}$ be the module of p -invariant measures λ such that $\lambda(\mathbb{X}'_p) = 0$.

Definition 6.1. Define the multiplicative van de Put lift

$$\text{ST}^{\times} : \mathbf{D}_0(\mathbb{Y}_p, \mathbb{Z})^{(0)} \longrightarrow \mathcal{A}^{\times}/p^{\mathbb{Z}}, \quad \lambda \longmapsto \left([\tau] \longmapsto \int_{\mathbb{X}_p} \tau^t \cdot x d\lambda(x) \right).$$

Remark 6.2. Note that we used that for $\lambda \in \mathbf{D}_0(\mathbb{Y}_p, \mathbb{Z})^{(0)}$, we have $\lambda(\mathbb{X}'_p) = 0$ to ensure that the integral defining $\text{ST}^{\times}(\lambda)([\tau])$ is independent of the choice of representative of $[\tau] \in \mathbb{P}^{n-1}(\mathbb{C}_p)$.

Proposition 6.3. *The morphism $\text{ST}^{\times}$ is $\text{GL}_n(\mathbb{Q}_p)$ -equivariant.*

Proof. The proof is similar to [DPV2, §1.4]. We begin with the following two observations. First, note that if h is a function on $\mathbb{Y}_p$ that can be integrated on $\mathbb{X}'_p$ and $\lambda \in \mathbf{D}_0(\mathbb{Y}_p, \mathbb{Z})$, we have

$$\oint_{\mathbb{X}'_p} h(x) d(g \cdot \lambda)(x) = \oint_{g^t \mathbb{X}'_p} (g^{-1} \cdot h)(x) d\lambda(x). \quad (9)$$

Second, note that for any $\lambda \in \mathbf{D}_0(\mathbb{Y}_p, \mathbb{Z})^{(p)}$ and $\gamma \in \mathrm{GL}_n(\mathbb{Q}_p)$, we have

$$\int_{\gamma \mathbb{X}'_p} \tau^t \cdot x d\lambda(x) \equiv \int_{\mathbb{X}'_p} \tau^t \cdot x d\lambda(x) \pmod{p^{\mathbb{Z}}}. \quad (10)$$

Indeed, there exist $a, b \in \mathbb{Z}$ for which

$$p^a \mathbb{X}'_p \subseteq \gamma \mathbb{X}'_p \subseteq p^b \mathbb{X}_p,$$

so we can find a decomposition into open sets

$$\mathbb{X}'_p = U_1 \sqcup \dots \sqcup U_t$$

for which

$$\gamma \mathbb{X}'_p = p^{r_1} U_1 \sqcup \dots \sqcup p^{r_t} U_t$$

for some integers $r_1, \dots, r_t$. Then, for each $1 \leq i \leq t$, we can use that λ is p -invariant to deduce

$$\int_{p^{r_i} U_i} \tau^t \cdot x d\lambda(x) \equiv \int_{U_i} \tau^t \cdot x d\lambda(x) \pmod{p^{\mathbb{Z}}},$$

which implies the desired congruence (10).

Now, we proceed to verify the desired equivariance of $\mathrm{ST}^\times$. Let $g \in \mathrm{GL}_n(\mathbb{Q}_p)$, $\lambda \in \mathbf{D}_0(\mathbb{Y}_p, \mathbb{Z})$. We would like to see that $\mathrm{ST}^\times(g \cdot \lambda) = (g \cdot \mathrm{ST}^\times(\lambda))$ in $\mathcal{A}^\times/p^{\mathbb{Z}}$

$$\mathrm{ST}^\times(g \cdot \lambda)([\tau]) = (g \cdot \mathrm{ST}^\times(\lambda))([\tau]).$$

For that, observe that for every $[\tau] \in \mathcal{X}_p$,

$$\begin{aligned} \mathrm{ST}^\times(g \cdot \lambda)([\tau]) &= \int_{\mathbb{X}'_p} (\tau^t \cdot x) d(g \cdot \lambda)(x) = \int_{g^t \mathbb{X}'_p} ((g^{-1}\tau)^t \cdot x) d\lambda(x) \\ &\equiv \int_{\mathbb{X}'_p} ((g^{-1}\tau) \cdot x) d\lambda(x) = (g \cdot \mathrm{ST}^\times)([\tau]) \pmod{p^{\mathbb{Z}}}, \end{aligned}$$

where we used (9) in the second equality for $h(x) = \tau^t \cdot x$, and (10) in the congruence modulo $p^{\mathbb{Z}}$. This implies that the quotient $\mathrm{ST}^\times(g \cdot \lambda)/(g \cdot \mathrm{ST}^\times(\lambda))$ is an invertible rigid analytic function taking values in $p^{\mathbb{Z}}$. By discreteness of $p^{\mathbb{Z}}$, we deduce that the quotient has to be a constant in $p^{\mathbb{Z}}$ which leads to the desired result. $\square$

6.2. Construction of rigid analytic classes. Recall the class

$${}_c\Theta^{\mathbf{D}} \in H^{n-1}(\mathrm{GL}_n^+(\mathbb{Z}_{(c)}), \mathbf{D}(\mathbb{X}, \mathbb{Z})^{(0)}/\delta_0).$$

Let $f \in \mathbb{Z}_{\geq 1}$ such that $p \equiv 1 \pmod{f}$ and let $\Gamma_1^p(f)$ be the p -arithmetic group of level f . Denote by $S_{\mathbb{Q}} := \{p\} \cup \{\ell \mid f\}$, where the second union runs over primes dividing f , and consider

$$V_f := e_1 + f \prod_{q \mid f} \mathbb{Z}_q^n \subset \prod_{q \mid f} \mathbb{Z}_q^n.$$

We can then consider the following map of measures

$$\mathrm{sp}_f: \mathbf{D}(\mathbb{X}, \mathbb{Z})^{(0)}/\delta_0 \longrightarrow \mathbf{D}(\mathbb{X}_{S_{\mathbb{Q}}}, \mathbb{Z})^{(0)}/\delta_0 \longrightarrow \mathbf{D}_0(\mathbb{X}'_p \times C_f, \mathbb{Z}) \longrightarrow \mathbf{D}_0(\mathbb{X}'_p, \mathbb{Z}) \simeq \mathbf{D}_0(\mathbb{Y}_p, \mathbb{Z})^{(0)},$$

where the first map is pushforward, the second restriction, and the third pushforward.

Lemma 6.4. *The previous map is $\Gamma_1^p(f)$ -equivariant.*

In view of the previous lemma, and using also the Van der Put lift $\text{ST}^\times$, we can define the following rigid analytic classes from ${}_c\Theta^\mathbf{D}$.

Definition 6.5. Define

$$J_f \in H^{n-1}(\Gamma_1^p(f), \mathcal{A}^\times/p^\mathbb{Z})$$

as $J_f := (\text{ST}^\times \circ (\text{sp}_f))_*({}_c\Theta^\mathbf{D})$.

Remark 6.6. When $n = 2$ and $f = 1$, then $\Gamma_1^p(f) = \text{SL}_2(\mathbb{Z})$. In this case, it can be verified that

$$J_f = J_{\text{DR}} \in H^1(\text{SL}_2(\mathbb{Z}[1/p]), \mathcal{A}^\times/p^\mathbb{Z})$$

is the rigid analytic class constructed by Darmon–Pozzi–Vonk in [DPV2, §2].

Remark 6.7. As in [DPV2, Section 2], the class J_f depends on the choice of c . However, that this dependency is quite mild. Indeed, if we denote by ${}_cJ_f$ the class J_f corresponding to J_f , we see that for any c and d primes different from p , we have

$$(1 - d^n){}_cJ_f = (1 - c^n){}_dJ_f.$$

In particular, $(1 - c^n)^{-1}{}_cJ_f$ is independent of c . We will continue working with a fixed c and omit the pre-script c as implicit.

6.3. Special points attached to totally real fields. Let F be a totally real field where p is inert. Denote by $\mathcal{O}$ its ring of integers.

Let $\mathfrak{a}$ be a fractional ideal of F prime to pf , and consider $\tau \in F^n$ a positively oriented $\mathbb{Z}$ -basis of $\mathfrak{a}^{-1}$ such that the isomorphism

$$\beta_\tau: \mathbb{Z}^n \xrightarrow{\sim} \mathfrak{a}^{-1}, \quad x \mapsto \tau^t \cdot x$$

satisfies $\beta(e_1) \equiv 1 \pmod{f_v \mathfrak{a}_v^{-1}}$ for every $v \mid f$. Recall that β_τ induces the embedding

$$F \hookrightarrow M_n(\mathbb{Q}), \quad \alpha \mapsto A_\alpha,$$

where A_α is determined as follows: for every $x \in \mathbb{Q}^n$, $\alpha(\tau^t \cdot x) = \tau^t \cdot A_\alpha x$. Finally, let

$$\rho_\tau: U_F(f) \hookrightarrow \text{SL}_n(\mathbb{Z}), \quad u \mapsto A_u^{-t}.$$

and observe that $\rho_\tau(U_F(f)) \subset \Gamma_1(f)$.

Lemma 6.8. *Via the fixed embedding $\bar{\mathbb{Q}} \subset \mathbb{C}_p$, view $\tau \in \mathbb{C}_p^n$. The element $[\tau] \in \mathbb{P}^{n-1}(\mathbb{C}_p)$ belongs to $\mathcal{X}_p$. Moreover, $[\tau]$ is fixed by $\rho_\tau(U_F(f))$.*

Consider the evaluation at τ map

$$\text{ev}_\tau: \mathcal{A}^\times/p^\mathbb{Z} \longrightarrow \mathbb{C}_p^\times/p^\mathbb{Z}.$$

Observe that this map is $\rho_\tau(U_F(f))$ -equivariant. Let $c_{U_F(f)} \in H_{n-1}(U_F(f), \mathbb{Z})$ be the generator corresponding to the positively oriented basis.

Definition 6.9. Consider the same notation as above. Let

$$J_f[\tau] := \text{ev}_\tau(J_\tau) \frown \rho(c_{U_F(f)}) \in \mathbb{C}_p^\times/p^\mathbb{Z}.$$

Then, as a consequence of the discussion of Section 5, we obtain the algebraicity of the values J_f at special points attached to totally real fields, which is the main result of the paper.

Theorem 6.10. *Consider the rigid analytic class $J_f \in H^{n-1}(\Gamma_1^p(f), \mathcal{A}^\times/p^\mathbb{Z})$ defined above. Let $\mathfrak{a}$ be a fractional ideal of F prime to pf , and consider $\tau \in F^n$ giving a basis of $\mathfrak{a}^{-1}$ as above. Let $H_{\mathfrak{f}}$ be the narrow ray class field of F of modulus $f\mathcal{O}_F$ and consider $u_S^{\sigma_{\mathfrak{a}}}$ the Gross–Stark unit determined in Theorem 5.1. Then,*

$$J_f[\tau] = u_S^{\sigma_{\mathfrak{a}}} \pmod{p^\mathbb{Z}}.$$

In particular, the value $J_f[\tau]$ is algebraic and belongs to the abelian extension $H_{\mathfrak{f}}$ of F .

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